

Tax Evasion and Productivity: Identifying and Correcting for Cost Overreporting*

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September 30, 2026

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Abstract

Value-added taxes (VAT) allow firms to deduct the taxes paid on purchases from the taxes collected on sales, creating an incentive for firms to overreport their purchases of inputs to reduce their tax bill. The fundamental issue in identifying input overreporting stems from two unobservables, true inputs and productivity. Because firms differ in productivity, a firm reporting high input use for a given level of output may be overreporting or less productive. I use a structural production-function approach and a subset of firms that share the same technology as the rest but report truthfully. I then use the common technology to predict how potentially tax-evading firms would behave without overreporting, and compare the prediction to what they report. My method requires only firm-level input and output data, and a truth-reporting group. The method thus applies equally to standard firm-level surveys, administrative data, or confidential tax records. I apply the method to a Colombian manufacturing survey from 1981 to 1991. During this period, the Colombian sales tax worked as a VAT and, due to government and market scrutiny, corporations refrained from overreporting inputs.

The approach yields a simple statistical test to detect input overreporting. Under the null hypothesis of no overreporting, the test is robust to a wide family of specifications of the probability of detection and the cost of overreporting. Among the 5 largest industries, the test detects overreporting in all 4 industries subject to the sales tax, and, as expected, fails to reject in the industry exempt from it. In the median of the 7 largest industries where overreporting is detected, firms report inputs that are 16 percent larger, on average, than true inputs. Next, I show that failing to correct for overreporting yields output elasticities of materials that are larger, and productivity distributions that are more dispersed and less persistent. I then estimate a structural model of firm tax evasion to study how claimed deductions, and thus government tax revenue, respond to a counterfactual tax rate. I find that a 0.5 percent increase in the sales-tax rate raises claimed deductions by 16 to 28 percent.

Finally, I exploit a sales-tax increase introduced by the 1983 reform to investigate how firms' overreporting responds to changes in the tax rate. I find that between 1983 and 1987, among the industries subject to the tax change, the ratio of overreported to true inputs rose by about 9 percentage points.

*I thank Salvador Navarro, David Rivers, and Timothy Conley for helpful comments. All errors are my own.

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JEL Classification: H26, D22, L60, O47

Keywords: Tax Evasion, Cost Overreporting, Production Function Estimation, Productivity

Introduction

Firms owned by the highest earners evade more tax through cost overreporting than firms owned by lower-income earners: relative to the bottom 80 percent of the income distribution, the tax evaded in Ecuador by the top 5 percent, as a share of owners' income, is 56 times larger (Carrillo et al. 2022). Despite its relevance, we still do not know much about firms' cost overreporting. Tax authorities around the world document this kind of tax evasion, often through fictitious invoices (OECD 2017), and the losses they report are significant: between 1.2 and 2.7 percent of total tax revenue in Chile, Colombia, Ecuador, Poland, and Slovakia, and about a ninth of the entire VAT compliance gap in Colombia and Slovakia (OECD 2017; Carrillo et al. 2022; CASE and IHS 2018; Longinotti 2022). Because some of these figures are tax administrations' own estimates, and all are conditional on detected evasion, true losses are plausibly larger.

Research has documented how individuals' reported income responds to tax rates, but has overlooked how firms' overreporting of inputs responds. For individuals, a large literature estimates how reported income responds to the tax rate and shows that the response includes evasion and avoidance as well as changes in real activity (Saez et al. 2012; Chetty 2009). For firms, evidence on responses to tax rates comes mostly from bunching at kinks and notches in the tax schedule (Best et al. 2015; Almunia and Lopez-Rodriguez 2018) and from misreported imports (Fisman and Wei 2004). Moreover, economic theory suggests that optimal commodity-tax rates should be lower on goods whose evasion is more elastic with respect to the tax rate (Cremer and Gahvari 1993). Yet, to my knowledge, no study has estimated how firms' overreporting of inputs, a form of tax evasion that the credit on purchases in value-added taxes incentivizes, responds to the tax rate. One reason is that true inputs are unobserved: for a given level of output, a firm reporting high input use may be overreporting, or it may suffer from low productivity.

I investigate whether input overreporting rises with the tax rate, and the extent to which it affects government tax revenue through firms' claimed deductions. Colombia's sales tax worked as a value-added tax, and in 1983 it raised the rate for most manufacturing industries but left a few exempt. I find that, between 1983 and 1987, overreporting by unincorporated firms in the affected industries rose significantly, by about 9 percentage points of true materials. In a structural model estimated on all industries, I find that a 0.5 percent increase in the sales-tax rate raises claimed deductions by 16 to 28 percent, while cuts lower claimed deductions only gradually: the smallest cut distinguishable from current policy is 8 percent. A back-of-the-envelope calculation suggests that, as a share of potential net sales-tax revenue from unincorporated firms in the sample, undetected overreporting costs about 3 percent. Near the current rate, and holding the tax collected on sales

fixed, the elasticity from the left of this revenue with respect to the tax rate is about -9 (Section A.5).

To distinguish between evasion and productivity, I employ a structural production-function approach, and then use it to study how evasion responds to the tax rate. In the absence of evasion, the first-order condition for materials, the flexible deductible input, reveals the structural relationship between the firm's materials share of revenue and the common technology within the industry. The government and market scrutiny keep corporations from overreporting. Corporations are thus the benchmark that allows me to identify the common technology in each industry and the distribution of measurement error in output. Among unincorporated firms, deviations from the common technology reveal overreporting up to measurement error in output. Deconvolution then separates the distribution of overreporting from that of the measurement error, and the production-function estimates recover productivity net of overreporting. To study how overreporting responds to the tax rate, I exploit the 1983 reform, measuring overreporting against the output elasticity of materials identified from corporations. Finally, employing an entropy-based method that does not require specifying the distribution of the unobservables (Schennach 2014), I estimate a structural model in which firms choose inputs and overreporting to maximize expected after-tax profits, weighing the tax evaded against the probability of detection and a cost of evasion that varies with productivity.

Separating overreporting from productivity also yields three findings of its own. First, a simple test compares unincorporated firms' materials shares with the corporate benchmark in each industry: under the null hypothesis of no overreporting, both should be equal; under the alternative, overreporting, the difference will be positive. Among the 20 largest industries, the test detects overreporting in 9 of the 18 industries liable for the sales tax, including 4 of the 5 largest, which together produce about two fifths of manufacturing output. It rejects decisively, at the 1 percent level, in 7 of them, where, in the median industry, unincorporated firms overreport an average of 16 percent of their true materials. It fails to reject in either of the 2 food industries, which were exempt from the sales tax and thus incentives to overreport were negligible. The test is valid for a broad class of models of detection and cost of evasion, and holds under any common production technology. A rejection is also unambiguous, because no firm gains by underreporting deductible inputs, so it can only point to overreporting. Second, failing to correct for overreporting biases estimates of the production function and of productivity. In those 7 industries, the uncorrected estimates (Gandhi et al. 2020) show output elasticities of materials that are consistently larger, and productivity distributions that are more dispersed and less persistent, than the corrected ones: in the median industry, the ratio of the 90th to the 10th percentile is 66 percent larger and the half-life of productivity shocks is 52 percent shorter. Third, the structural estimates show a cost of evasion that is convex in productivity and lowest near the 86th percentile of productivity, which rationalizes a pattern observed in other studies: evasion rises with firm size but drastically drops at the very top (Carrillo et al. 2022).

These results relate to three strands of public economics and to one of productivity measurement. First, I estimate how a particular form of evasion by firms responds to the tax rate, a response

that the literature on taxes and revenue has mostly studied through individuals' reported income. Traditionally, the elasticity of personal taxable income has combined all the different responses to the rate, evasion included, into a single statistic (Feldstein 1999; Slemrod and Kopczuk 2002; Saez et al. 2012). However, when part of the cost of evasion is a transfer rather than a resource cost, this elasticity alone no longer measures the efficiency cost of taxation (Chetty 2009). Measuring that cost requires knowing how evasion itself responds to the rate, separately from real activity. Direct evidence on this response is scarcer, and comes from the gap between households' consumption and reported income (Gorodnichenko et al. 2009) and, for firms, from bunching at kinks and notches in the tax schedule (Best et al. 2015; Almunia and Lopez-Rodriguez 2018). I provide evidence for firms' overreporting of inputs under a sales tax that credits purchases, from both a statutory rate change and a structural model that separates overreporting from productivity. The structural model also speaks to the theory of optimal commodity taxation under evasion by estimating how claimed deductions respond to the rate (Cremer and Gahvari 1993).

Second, I provide measures of firms' cost overreporting without relying on confidential tax records. Most evidence on firm evasion focuses on underreported revenue (Slemrod 2019), yet the cost side matters. Firms offset enforcement on revenue by raising reported costs (Carrillo et al. 2017; Asatryan and Peichl 2017), and paper trails make transactions between firms harder to misreport than sales to final consumers (Pomeranz 2015); therefore, firms that sell to other firms have little room to misreport on the sales side and more on the cost side. Cost overreporting has mostly been measured directly with transaction-level tax records in Ecuador and Mexico. These studies reveal the mechanism: firms acquiring fake invoices from ghost firms (Carrillo et al. 2022; Zumaya et al. 2021). Tax records, however, are confidential, available for few countries, and accessible only to selected researchers. Survey-based measures of firm evasion rely instead on firms' self-reported share of sales declared to the tax authority (Dabla-Norris et al. 2019), whose truthfulness cannot be verified. My approach needs only reported inputs and output, and is thus equally applicable to standard firm-level surveys, administrative records, and confidential tax records.

Third, I extend the validation-sample approach from individuals to firms. This approach infers evasion by comparing an indirect measure of true income between a truth-reporting group and a group that may misreport. For example, the self-employed report less income relative to their food consumption than employees, whose income is reported by their employers (Pissarides and Weber 1989). Researchers have also used the same comparison to measure how evasion responds to a tax reform (Gorodnichenko et al. 2009). For firms, however, the natural indirect measure, output relative to inputs, also depends on an additional unobservable, productivity. A firm reporting high input use may be overreporting or may simply have a negative productivity shock. I use the production function to separate the two, with corporations as the truth-reporting group. Unlike the comparisons for individuals, my approach does not require the two groups to be alike in size or productivity: corporations are much larger than unincorporated firms, and the production function accounts for the differences in productivity behind this gap.

Finally, the paper shows how tax-motivated overreporting of inputs biases production-function estimates. Proxy-variable methods rely on a flexible input, typically materials, either inverting its demand (Levinsohn and Petrin 2003; Akerberg et al. 2015) or using its first-order condition (Gandhi et al. 2020). These methods treat reported inputs as true. Work on mismeasured inputs has focused on classical measurement error (mean zero and independent of the true input), mostly in capital (Collard-Wexler and De Loecker 2020). Overreporting is neither: firms choose it based on their own productivity. Furthermore, it affects exactly the input on which these methods rely. Failing to correct for overreporting yields output elasticities of materials that are consistently larger, and productivity distributions that are more dispersed and less persistent, than the corrected estimates. These findings matter for any analysis built on measured productivity differences across firms.

The rest of the paper follows the argument. Section 1 shows how the tax rate generates incentives for a firm to overreport materials, and how its optimal choice depends on the probability of detection and the cost of evasion. Section 2 explains why Colombian corporations can serve as truth-tellers. Section 3 shows how overreporting can be detected and measured from production data alone. Then, Section 4 demonstrates how failing to correct for overreporting biases production-function and productivity estimates. Section 5 leverages the 1983 reform to document that overreporting responds to the tax rate, and Section 6 turns the question around and asks how much revenue the government would lose to overreporting as it adjusts the tax rate. Section 7, as tradition demands, concludes.

1 A Model of Tax Evasion through Input Overreporting

This chapter models a firm that overreports its materials to reduce its sales tax liability. The sales tax worked as a VAT. The firm credits the tax paid on its purchases against the tax it owes on its sales, so every peso of fictitious materials lowers its tax bill by the rate on purchases. The probability of being caught and the resources allocated to concealment weigh against this benefit. Colombian manufacturers typically faced different rates on their sales and on their purchases (Section 2.2). In 56 percent of firm-years the two rates differ by more than one percentage point. I therefore allow firms to have two rates. This is not needed for identification: only the rate on purchases enters the decision to overreport, so a single-tax model would do just as well.

1.1 Setup

A firm i in year t produces output with capital K_{it} , labour L_{it} , and materials M_{it} ,

$$Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it}), \quad (1)$$

where ω_{it} is productivity, known to the firm when it chooses its inputs, and ε_{it} is measurement error in output, with $\mathbb{E}[\varepsilon_{it}] = 0$. Because ε_{it} is an error in the record of output, not a shock to production, the firm's decisions depend on true output, $K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it})$, and productivity is

$\exp(\omega_{it})$. In Gandhi et al. (2020), by contrast, ε_{it} is a shock to output that the firm does not know when it chooses its inputs. I follow the production-function literature in treating materials as the flexible input, and capital and labour as predetermined or dynamic.

The firm sells its output at price P_t and owes sales tax at rate τ_S on its sales. It buys materials at price ρ_t and credits, at rate τ_P , the tax paid on them against the tax it owes; when the credit exceeds the tax owed, as for firms exempt from the tax on their sales, the tax authority refunds the difference. The two rates vary across firms and years, because firms sell and buy goods taxed at different rates. The firm claims the credit on reported purchases, $M_{it}^* = M_{it} + e_{it}$, where $e_{it} \geq 0$ denotes overreported materials. Capital goods and labour carry no credit (Section 2.2), so the firm overreports only materials.

Overreporting carries two costs. First, following Allingham and Sandmo (1972), the tax authority audits the firm with probability $q(e_{it})$, which increases with overreporting, with $q(0) = 0$, and an audit always uncovers the overreported materials. A firm that is caught repays the credit claimed on e_{it} plus a fine, a multiple ϕ of that credit. Second, concealment diverts resources, such as the manager's time, from production; $\kappa_{it}(e_{it}, \omega_{it})$ denotes this cost of evasion, which increases in e_{it} . Neither the detection probability nor the cost of evasion depends on true materials, so the firm decides how much to overreport separately from how much material to use.

The firm chooses its inputs and overreporting to maximize expected profit,

$$\begin{aligned} \max_{K,L,M,e} \mathbb{E}[\pi_{it}] = & (1 - \tau_S)P_t K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it}) - (1 - \tau_P)\rho_t M_{it} - r_t K_{it} - w_t L_{it} \\ & + \tau_P \rho_t e_{it} [1 - (1 + \phi)q(e_{it})] - \kappa_{it}(e_{it}, \omega_{it}). \end{aligned} \quad (2)$$

The first term is revenue net of the tax on sales; the second, the cost of true materials net of the credit on them; the third and fourth, the rental cost of capital and the wage bill. The fifth term is the credit on overreported materials that the firm expects to keep, net of the expected repayment and fine, and the last is the cost of evasion.

Setting $\tau_S = \tau_P = \tau$ in Equation 2 recovers the single-tax profit of Section B.1.

1.2 First-order conditions

The firm uses materials until their marginal revenue, net of the tax on sales, equals their price, net of the credit:

$$(1 - \tau_S)P_t \beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta-1} \exp(\omega_{it}) = (1 - \tau_P)\rho_t. \quad (3)$$

Overreporting does not enter Equation 3, because the credit on e_{it} does not depend on true materials. The two rates enter only through the ratio $(1 - \tau_P)/(1 - \tau_S)$: when the rate on purchases exceeds the rate on sales, the credit lowers the price of materials relative to output, and the firm uses more of them.

Multiplying Equation 3 by M_{it} and dividing by measured output, $P_t Y_{it}$, gives the log share of materials in output, net of sales taxes:

$$\ln \left(\frac{(1 - \tau_P) \rho_t M_{it}}{(1 - \tau_S) P_t Y_{it}} \right) = \ln \beta - \varepsilon_{it}. \quad (4)$$

With a single rate, the tax terms cancel and the gross share, $\rho_t M_{it} / P_t Y_{it}$, satisfies the same equation. Section 3 builds on Equation 4 to identify β .¹

The firm overreports until the credit it expects to keep on the marginal peso equals the marginal cost of evasion. For an interior choice,

$$\tau_P \rho_t [1 - (1 + \phi)(q(e_{it}) + q'(e_{it})e_{it})] = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}}. \quad (5)$$

Only the rate on purchases enters Equation 5, because the tax on sales does not depend on reported materials. A higher rate on purchases raises the credit on every fictitious peso, whether or not the firm is caught, so the firm overreports more. By the implicit function theorem, $\partial e_{it}^* / \partial \tau_P > 0$ for any detection probability and cost of evasion that satisfy the second-order condition below.

Some firms choose not to overreport at all. For them, the marginal cost of evasion at zero already exceeds the marginal credit, $\tau_P \rho_t \leq \partial \kappa_{it}(0, \omega_{it}) / \partial e_{it}$, and Equation 5 holds with $\leq$ instead of $=$. Firms that pay no tax on their purchases ($\tau_P = 0$) are always in this group.

Corporations are different. Audited statements and scrutiny by government and markets keep them from overreporting (Section 2.3), so $e_{it} = 0$ is a constraint for them, not an outcome of Equation 5. I use them only as the truth-reporting benchmark in Section 3.

1.3 Second-order condition

Profit's Hessian is block-diagonal: $M_{it} \perp e_{it}$, so no cross-partial links overreporting to capital, labour, or materials, and the production function is strictly concave in (K_{it}, L_{it}, M_{it}) . The second-order condition therefore reduces to a single scalar condition, $\partial^2 \mathbb{E}[\pi_{it}] / \partial e_{it}^2 < 0$. With a cost of evasion linear in e_{it} and $\phi = 0$, this condition depends only on the curvature of the detection probability (derivation in Section B):

$$-\frac{q''(e_{it})}{q'(e_{it})} e_{it} < 2. \quad (6)$$

The detection probability may be concave, but not too concave: the semi-elasticity of its marginal probability q' with respect to e_{it} must stay above -2 . A linear q satisfies the condition everywhere, since $q'' = 0$, as does any convex q ; a concave q such as $1 - e^{-\lambda e_{it}}$ satisfies it only when $\lambda e_{it} < 2$.

¹The two shares differ little in the data. The log ratio $\ln((1 - \tau_S)/(1 - \tau_P))$ has a standard deviation of 0.05 across firm-years, against 0.34 for the log gross share.

2 Setting and Data

Colombian manufacturing in 1981–1991 offers a unique setting to study tax evasion by overreporting deductible inputs. First, a sales tax that worked as a value-added tax and rewarded overreporting materials: manufacturers could reduce the tax they owed on their sales by claiming fictitious deductions (Section 2.2). Second, a truth-reporting group of firms: corporations, which government and market scrutiny kept from evading (Section 2.3). Third, variation in the tax rate: the 1983 reform raised the sales-tax rate for most industries but left food products exempt (Section 2.4). Finally, firm-level records of output and inputs, together with legal form and taxes paid (Section 2.1). Because the approach uses the production function, it applies to any data with firms’ output and inputs, such as surveys, administrative data, or tax records.

2.1 Data

I use the Colombian Annual Survey of Manufacturing (EAM), a panel of manufacturing plants with more than 10 employees from 1981 to 1991.² The EAM records, for each plant, the output and inputs that a production function needs, the plant’s juridical organization, and the sales tax it charged on its sales and paid on its purchases.³ The panel is well known: it has been used in seminal work on export dynamics, reallocation, and productivity, and to estimate production functions (Roberts and Tybout 1997; Das et al. 2007; Eslava et al. 2004; Gandhi et al. 2020). Similar to these studies, I observe inputs as firms reported them; in contrast, I ask how the tax incentives of the period, including the 1983 reform, shaped those reports. The working sample has 40,508 firm-years from 5,915 plants in 28 industries, covering the four juridical organizations described in Section 2.3 (Table 1).

2.2 The sales tax: a credit on purchases

The Colombian sales tax worked as a value-added tax (VAT): manufacturers credited the tax paid on their purchases against the tax charged on their sales. The tax originally targeted the manufacturing sector, on finished goods and imports. From 1966, manufacturers could claim credit on the taxes paid on inputs *incorporated* into the product, and from 1974 on any purchase made by the firm, except the acquisition of capital goods; the credits worked through a system of refunds (Perry and Orozco de Triana 1990, 180).

The credit rewards overreporting materials. An artificial increase in the reported value of materials

²The survey’s unit of observation is the plant; I refer to plants and firms interchangeably.

³The year in the panel is the activity (reference) year. DANE runs the EAM each year with the economic year immediately preceding collection as the reference period (Departamento Administrativo Nacional de Estadística (DANE) 1995, 40): the survey for year x is collected during $x + 1$ and labelled x (e.g. the 1992–1994 surveys were collected in 1993–1995, and the 1992 file values assets at December 1992; Departamento Administrativo Nacional de Estadística (DANE) (2018)).

Table 1: Summary statistics, manufacturing firms in Colombia (1981–1991).

Variable	Missing (%)	Mean	SD	Q1	Median	Q3
Log gross output, y	0.00	10.53	1.78	9.18	10.28	11.69
Log capital, k	0.00	9.03	2.02	7.69	8.94	10.34
Log labour, l	0.00	3.84	1.26	2.89	3.68	4.64
Log raw materials, m^*	0.57	9.61	2.05	8.24	9.46	10.96
Sales-tax rate on sales, τ_S	0.06	0.066	0.049	0.007	0.070	0.100
Sales-tax rate on purchases, τ_P	0.64	0.074	0.059	0.027	0.077	0.105

Firm-years with finite output, capital, labour and materials; corporations, LLCs, partnerships and proprietorships. τ_S : sales tax paid on sales over sales; τ_P : sales tax paid on purchases over raw materials. Values above 1 are treated as data errors and counted as missing. Gross output, capital and raw materials are deflated to 1981 prices; labour is in employee-years.

raises the credit, and so lowers the firm’s tax liability, by the tax rate paid on purchases, peso per peso. Capital goods could not be credited and labour was not a taxed purchase, so I focus on raw materials consumed.⁴

Weak enforcement invited firms to collect this reward. Tax evasion flourished in Colombia during the sample period (Sánchez Torres and Gutiérrez Sourdis 1994), while the tax authority audited inefficiently, strained under its workload, and contended with legal loopholes (Perry and Orozco de Triana 1990).

2.3 Juridical organizations: corporations as truth reporters

I treat corporations as truthful reporters of materials because two kinds of scrutiny restrained them from overreporting: government and market. First, the Superintendent of Corporations closely supervised corporations and required each to have an auditor, according to the OAS/IDB Joint Tax Program’s *Fiscal Survey of Colombia* (1965, 27–30); LLCs and proprietorships faced only the tax authority, whose audits were weak. Second, tradable shares subject corporations to market discipline, and the market penalizes aggressive tax behaviour even when it is legal. In the United States, stock prices fall, on average, when news of a corporation’s involvement in tax shelters becomes public, although sheltering is avoidance rather than evasion (Hanlon and Slemrod 2009; Slemrod 2019). If the market penalizes legal avoidance, it should penalize illegal overreporting at least as much. Evidence from Ecuador agrees: the tax evaded by claiming fictitious purchases amounted to 1.7 percent for corporations and 11.5 percent for sole proprietorships (Carrillo et al. 2022). If corporations nonetheless overreport, my estimates understate the overreporting of unincorporated

⁴I exclude the other intermediates. Energy, fuels, and repair and maintenance are small, with medians of 1.0 percent, 0.3 percent, and 0.3 percent of gross output, and fuels and repair and maintenance are reported as zero in 8 and 15 percent of firm-years. Services, a median of 8.5 percent of gross output, entered the tax only gradually: the 1974 reform extended it to some services, and the 1983 reform to others, such as hotels, computing, maintenance, and rentals (McLure 1989, 65).

Table 2: Juridical organization, manufacturing firms in Colombia (1981–1991).

J. Org.	Firm-years	Plants	% of firm-years
Proprietorship	5,343	1,226	13.2
LLC	25,456	3,927	62.8
Partnership	1,425	319	3.5
Corporation	8,284	1,003	20.5

A plant that changes juridical organization is counted once in each.

firms; they are a lower bound.

The data distinguish several juridical organizations. Four of them account for 98 percent of firm-years and 95 percent of real gross output⁵: corporations, limited liability companies (LLCs), partnerships, and proprietorships (Table 2). I define them following the *Fiscal Survey of Colombia* (1965, 27–30) and DANE’s documentation of the EAM (2018, 12). Corporations (*sociedades anónimas*) are associations of capital, the counterpart of the US corporation: at least five shareholders, tradable shares of equal value, and liability limited to the capital contributed (1965; 2018). An LLC (*sociedad de responsabilidad limitada*) has two or more partners, not exceeding 20 (1965) or, by the 1990s, 25 (2018), whose stakes (*cuotas*) cannot be traded and whose liability is limited to the capital contributed; it is an association of persons, not of capital, and is not subject to the Superintendent of Corporations unless more than one-third of it is owned by a corporation (1965). Partnerships are associations of two or more persons in which the partners, or at least the managing partners, are jointly, severally, and unlimitedly liable for the partnership’s operations (2018). They are a small group (3.5 percent of firm-years), so I pool them with the other unincorporated firms. Proprietorships are individuals who run the business in their own name, without a separate legal personality (2018).⁶

Because of the income tax, incentives to overreport varied within industries by juridical organization (Table 3). Firms could reduce their taxable profits and, with them, the income tax by deducting the cost of fictitious materials. Between 1974 and 1983, partnerships and LLCs paid a rate of 20 percent, and their owners paid individual income tax again on their share of the profits, whether distributed or not (McLure 1989, 67; Perry and Cárdenas 1986, 1:23). Proprietors paid only the individual income tax, on a graduated schedule with a top rate of 56 percent (Perry and Cárdenas 1986, 1:36). Corporations paid a rate of 40 percent between 1974 and 1986, and their shareholders paid individual income tax again only on the dividends they received (McLure 1989, 67; Perry and Cárdenas 1986, 1:23). Even though corporations had incentives to overreport too, the government and market scrutiny described above restrained them.

⁵The other organizations in the data are excluded: stock partnerships (*sociedades en comandita por acciones*), whose shares are tradable and which are taxed as corporations (Joint Tax Program of the Organization of American States and the Inter-American Development Bank 1965), and non-profit or state organizations, such as cooperatives, community enterprises, and state industrial enterprises, whose proceeds come from taxes, fees, or special contributions.

⁶The *empresa unipersonal*, a single-owner legal entity, was introduced only in 1995 (Congreso de la República de Colombia 1995, art. 71).

Table 3: Income tax, liability, capital, and owners by juridical organization (1974–1983).

Organization	Income tax (company)	Income tax (owners)	Liability	Capital	Owners
Corporation	40%	On dividends received	Limited to capital contributed	Tradable shares	$N \geq 5$
LLC	20%	On all profits	Limited to capital contributed	Non-tradable stakes (<i>cuotas</i>)	$2 \leq N \leq 20$ (25)
Partnership	20%	On all profits	Unlimited ^a	Not a capital association	$N \geq 2$
Proprietorship	–	Individual schedule	Unlimited	Owner's own assets	$N = 1$

Rules in force from the 1974 reform until Ley 9 de 1983, which cut the LLC rate to 18%. Company: entity-level income-tax rate. Owners: individual income tax; proprietors paid only the graduated individual schedule, with a top rate of 56%. The limit on LLC partners is 20 in the 1965 source and 25 in the 1992–1994 survey documentation. Sources: tax rates, McLure (1989), p. 67, and Perry and Cárdenas (1986), vol. 1, pp. 23 and 36; liability, capital and owners, Fiscal Survey of Colombia (1965), pp. 27–30, and DANE (2018), p. 12.

^a All partners in general partnerships; the managing partners in ordinary limited partnerships.

2.4 The 1983 reform

The 1983 reform altered the reward to overreporting across and within industries. Across industries, the reform raised the sales-tax rate in most liable industries and left exempt industries untouched. Within industries, it cut the income-tax rates by different amounts for each juridical organization: most for proprietorships, less for LLCs, and not at all for corporations. For unincorporated firms in liable industries, the two changes pulled in opposite directions: the higher sales tax raised the reward to overreporting, and the lower income tax reduced it. I exploit this variation in Section 5 to test whether overreporting responded to its reward.

On the sales tax, the reform combined the items previously taxed at 6 and 15 percent into a single 10 percent bracket (Perry and Orozco de Triana 1990, 182), while food products (industries 311 and 312), exempt since the tax was introduced (Perry and Orozco de Triana 1990, 181), remained exempt. The typical taxed firm was in the 6 percent bracket, so the reform raised its rate: between 1983 and 1985, the median rate on sales rose from 6 to 10 percent, and the median rate on purchases, the rate at which each fictitious peso of materials was credited, from 6.6 to 9.9 percent (Table 4).

The new rate, set by Decreto 3541 of 29 December 1983 under the extraordinary powers of Ley 9 de 1983, applied from 1 April 1984 (Presidencia de la República de Colombia 1983; Corte Constitucional de Colombia 2016; Perry and Orozco de Triana 1990), so 1983 is the last full pre-reform year and 1984 a partial one. The observed rates match this timing (Table 4). The median rate paid on sales is 6 percent in 1981–1983, 10 percent from 1985, and 12 percent in 1991, after Ley 49 de 1990 raised the general rate to 12 percent (Congreso de la República de Colombia 1990). In 1984 it is 9 percent, exactly what three months at 6 percent and nine months at 10 percent give ($0.25 \times 6\% + 0.75 \times 10\% = 9\%$).

Table 4: Observed sales-tax rates by survey year, on sales (τ_S) and on purchases (τ_P), against the statutory general rate in force. Rows in bold are the years a reform took effect: 1984 (Decreto 3541 de 1983, from 1 April) and 1991 (Ley 49 de 1990, from 1 January). Firms with a rate strictly between 0 and 50 percent.

Year	Statutory general rate	N	Median τ_S	Mean τ_S	P75 τ_S	Median τ_P
1981	15% basic, 6% preferential	4,399	6.0%	7.8%	9.8%	6.7%
1982	15% basic, 6% preferential	4,098	6.0%	7.7%	9.5%	6.1%
1983	15% basic, 6% preferential	3,273	6.0%	7.7%	9.5%	6.6%
1984	10% from 1 April	3,005	9.0%	8.2%	10.0%	8.7%
1985	10%	2,781	10.0%	9.0%	10.0%	9.9%
1986	10%	2,623	10.0%	9.0%	10.0%	9.8%
1987	10%	2,485	10.0%	8.9%	10.0%	9.9%
1988	10%	2,360	10.0%	8.9%	10.0%	9.8%
1989	10%	2,265	10.0%	8.9%	10.0%	9.7%
1990	10%	2,176	10.0%	8.8%	10.0%	9.7%
1991	12% from 1 January	2,063	12.0%	10.3%	12.0%	11.5%

τ_S : sales tax charged on sales over sales; N and the τ_S columns use firms with $0 < \tau_S < 50\%$ (ST-exempt firms excluded). τ_P : sales tax paid on purchases over raw materials, median over firms with $0 < \tau_P < 50\%$.

On the income tax, Ley 9 de 1983 cut the individual schedule, faced by proprietorships, by 4.6 percentage points on average, with the top rate falling from 56 to 49 percent, and the LLC rate from 20 to 18 percent (Perry and Cárdenas 1986, 1:36).

3 Identifying Tax Evasion

3.1 Non-evaders

Because the firms' optimization decisions depend on the fiscal environment, the identification strategy should be motivated by the fiscal environment Γ . In particular, the identification strategy will be as good as how well we can tell apart a subset of firms that have the highest incentive to not evade. For example, in the case of Spain, the firms above the revenue LTU threshold. In the case of Colombia, the corporations (Section 2.3).

Assumption 1 (Non-evaders). Based on the fiscal environment Γ , the researcher can identify a subset of firms $\theta_i \in \Theta^{NE} \subset \Theta$ that does not evade taxes by overreporting inputs.

For those firms, then $\mathbb{E}[u_{it} | \theta_i \in \Theta^{NE}] = 0$.

In addition, I impose the following timing assumption.

Assumption 2 (Independence). The measurement error in output, ε_{it} , is independent of overreporting u_{it} and of true materials M_{it} , with $\mathbb{E}[\varepsilon_{it}] = 0$.

3.2 Identifying the output elasticity of materials

Assume the production function is Cobb-Douglas, $M_{it}^\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} \exp(\omega_{it} + \varepsilon_{it})$. The econometrician observes output, capital, labour, and prices of intermediate inputs and output, and the sales-tax rates on sales and on purchases, $\{Y_{it}, K_{it}, L_{it}, \rho_t, P_t, \tau_S, \tau_P\}$, but she only observes overreported intermediate inputs, $M_{it}^* = M_{it} + e_{it}$. Let $u_{it} \equiv \ln(M_{it}^*/M_{it}) = \ln(1 + e_{it}/M_{it})$ denote the log overreporting ratio.

Let s_{it} denote the log share of reported materials in output, net of sales taxes, $s_{it} \equiv \ln((1 - \tau_P)\rho_t M_{it}^*/(1 - \tau_S)P_t Y_{it})$.⁷ Writing the net share of Equation 4 for true and for reported materials, we get

$$\ln\left(\frac{(1 - \tau_P)\rho_t M_{it}}{(1 - \tau_S)P_t Y_{it}}\right) = \ln \beta - \varepsilon_{it} \quad (7)$$

$$s_{it} = \ln \beta - \varepsilon_{it} + u_{it}$$

where Y_{it} is output as measured, with measurement error ε_{it} . We can use Equation 7 and Assumptions 1 and 2 to recover the output elasticity of materials, β ,

$$\ln \beta = \mathbb{E}[s_{it} \mid \Theta^{NE}]. \quad (8)$$

3.3 The observed residual

Having recovered the output elasticity of materials, β , for all firms, I can form the following variable using observed data.

$$\mathcal{V}_{it} \equiv s_{it} - \ln \beta = u_{it} - \varepsilon_{it} \quad (9)$$

Tax evasion therefore can be recovered up to an independent random variable.

3.4 Testing for overreporting

Equation 9 suggests a way to test the presence of tax evasion through cost overreporting. Let $\mathbb{E}[\mathcal{V}_{it}] \equiv \mu_{\mathcal{V}}$ and define the null hypothesis as the absence of cost overreporting and the alternative hypothesis as the presence of cost overreporting. Consequently, to verify for the presence of tax evasion by overreporting, we can use a one-sided test where under the alternative hypothesis $H_1 : \mu_{\mathcal{V}} > 0$, and under the null, $H_0 : \mu_{\mathcal{V}} \leq 0$.

⁷In the single-tax model (Section B.1) the tax terms cancel and s_{it} is the gross share, $\ln(\rho_t M_{it}^*/P_t Y_{it})$.

Table 5: Test for overreporting of materials by industry. Mean of $\mathcal{V}_{it}$ among unincorporated firms and 95 percent test-inversion region for its expectation (sharp test), with $\ln \beta$ estimated from corporations.

Industry	$\hat{\mu}$	95% region, sharp
311 Food products	0.024	($\cdot$, 0.048]
312 Other food products	-0.018	($\cdot$, 0.076]
313 Beverages	0.390	[0.310, 0.470]
321 Textiles	0.151	[0.115, 0.188]
322 Wearing apparel	0.224	[0.195, 0.253]
323 Leather products	-0.192	†
324 Footwear	0.095	[0.059, 0.131]
331 Wood products	0.250	[0.175, 0.325]
332 Furniture	-0.007	($\cdot$, 0.059]
341 Paper products	-0.066	($\cdot$, 0.004]
342 Printing and publishing	0.074	[0.036, 0.112]
351 Industrial chemicals	0.165	[0.001, 0.329]
352 Other chemicals	0.057	[0.002, 0.111]
356 Plastic products	-0.107	†
369 Non-metallic minerals	0.193	[0.109, 0.277]
381 Metal products	-0.005	($\cdot$, 0.025]
382 Non-electrical machinery	-0.014	($\cdot$, 0.026]
383 Electrical machinery	0.034	($\cdot$, 0.089]
384 Transport equipment	-0.129	†
390 Other manufacturing	0.057	($\cdot$, 0.115]

$\hat{\mu}$: mean of $\mathcal{V}_{it} = \ln(\rho_t M_{it}^* / P_t Y_{it}) - \ln \hat{D}$ among unincorporated firms, with the log materials share net of sales taxes and $\ln \hat{D}$ the mean among corporations. 95% test-inversion region for $\mu = E[\mathcal{V}_{it}]$ over the grid $\mu \in [0, 1]$ (under the model $\mu \geq 0$), taking $\ln \hat{D}$ as the true value; plant-clustered, efficiently weighted moments; sharp, $\chi^2_{1,0.95} = 3.84$ critical value (the corporations' moment holds exactly at $\ln \hat{D}$). Where the region excludes 0, the test rejects no overreporting. ($\cdot$, b): $\mu = 0$ is not rejected. † Empty region: every $\mu \in [0, 1]$ is rejected.

I invert the test. For each industry, Table 5 displays the estimated mean of $\mathcal{V}_{it}$ among unincorporated firms and the values of $\mu_{\mathcal{V}}$ in $[0, 1]$ that the test does not reject at the 5 percent level, taking $\ln \hat{\beta}$ from corporations as the true value. The test detects overreporting where this set excludes zero. Section D reports a conservative version of the test and a bootstrap test that carries the sampling error of $\ln \hat{\beta}$.

3.5 Deconvolving tax evasion

We can do better however. By Assumption 2, the measurement error ε_{it} is independent of overreporting u_{it} . Note that, from Equation 7, we also recovered $f_{\varepsilon}(\varepsilon)$, the distribution of ε . By using deconvolution methods, I can recover the tax evasion apart from the output shock.

In particular, from probability theory, the density of the sum of two **independent** random variables is equal to the **convolution** of the densities of both addends (Meister 2009). Hence, from Equation 9,

$$f_{\mathcal{V}}(\mathcal{V}) = \int f_u(\mathcal{V} + \varepsilon) f_\varepsilon(\varepsilon) d\varepsilon \quad (10)$$

where $f_{\mathcal{V}}$ denotes the density of $\mathcal{V}$.

3.6 Estimation

Following Kang and Qiu (2021), consider the following logspline model:

$$f_{u|\Theta}(u) = \exp(s(u; \theta) - C(\theta)), \quad L < u < R$$

where $[L, R]$ define the support of f_u ,

$$s(u; \theta) = \sum_{j=1}^{K-1} \theta_j B_j(u),$$

$\{B_j(u), j = 1, 2, \dots, K\}$ is a sequence of cubic B-spline basis functions, and

$$C(\theta) = \log \left(\int_L^R \exp(s(u; \theta)) du \right) < \infty.$$

Then $f_{u|\Theta}(u)$ is a positive density function on (L, R) . The logspline model assumes that f_u is a twice continuously differentiable function. The last basis function is not used for identifiability purposes, given that by the property of B-splines, $\sum_{j=1}^K B_j(u) = 1$ for all $u \in (L, R)$.

The log likelihood of the observed variable $\mathcal{V}$ is

$$\begin{aligned} l_{\mathcal{V}}() &= \sum_{i=1}^n \log \left(\int f_\varepsilon(u - \mathcal{V}_i) f_{u|\theta}(u) du \right) \\ &= \sum_{i=1}^n \log \left(\int f_\varepsilon(u - \mathcal{V}_i) \exp(s(u; \theta)) du \right) - nC(\theta) \end{aligned}$$

The usual maximum likelihood estimate $\hat{\theta}$ is the maximizer of $l_{\mathcal{V}}(\theta)$.

To avoid overfitting, I add a penalty term to the log-likelihood function. The penalty term is a linear combination of the squared second derivatives of the B-spline basis functions. The second derivative of the B-splines is approximated by finite differences (Eilers and Marx 1996). The penalty term is given by

Table 6: Distribution of the overreporting ratio e/M by industry, unincorporated firms.

Industry	Mean	SD	P10	Median	P90	$E[u]$	Test, $E[\mathcal{V}]$
313 Beverages	48.6%	3.8%	43.8%	48.6%	53.4%	0.396	[0.310, 0.470]
321 Textiles	16.3%	5.2%	9.7%	15.9%	23.3%	0.150	[0.115, 0.188]
322 Wearing apparel	23.1%	3.9%	18.0%	23.0%	28.1%	0.207	[0.195, 0.253]
324 Footwear	10.5%	5.8%	4.4%	9.3%	18.2%	0.098	[0.059, 0.131]
331 Wood products	24.7%	10.0%	13.1%	23.3%	38.4%	0.218	[0.175, 0.325]
342 Printing and publishing	4.3%	8.0%	2.9%	3.7%	4.6%	0.040	[0.036, 0.112]
369 Non-metallic minerals	10.4%	7.5%	6.8%	10.0%	13.8%	0.098	[0.109, 0.277]

Overreporting ratio $x = e/M$: overreported materials as a share of true materials. Moments and quantiles of the density $f_x(y) = f_u(\ln(1+y))/(1+y)$, obtained by transforming the deconvolved density of $u = \ln(1 + e/M)$ (penalized B-spline deconvolution, unincorporated firms, log materials share net of sales taxes). Industries: those whose 99% sharp test-inversion region for $E[\mathcal{V}]$ excludes zero, a rule fixed before deconvolving. $E[u]$: mean of the deconvolved u ; under the model it equals $E[\mathcal{V}]$, whose 95% sharp test-inversion region is shown for comparison.

$$\text{Penalty}(\theta) = \lambda \cdot \|\mathbf{D}\theta\|^2$$

Intuitively, we want to penalize excessive curvature in the log-density $\log f_u(u) \approx s(u; \theta) = \sum_j \theta_j B_j(u)$, where B_j are B-spline basis functions.

3.7 The overreporting ratio

The deconvolved object is the log ratio $u_{it} = \ln(1 + e_{it}/M_{it})$. The overreporting ratio itself, $x_{it} \equiv e_{it}/M_{it} = \exp(u_{it}) - 1$, is a monotonic, differentiable transformation of u_{it} , so its density follows from a density transformation (Hogg et al. 2019, Theorem 1.7.1),

$$f_x(y) = f_u(\ln(1+y)) \left| \frac{1}{1+y} \right|, \quad y \geq 0. \quad (11)$$

Table 6 and Figure 1 display the estimated distribution of the overreporting ratio by industry.

4 Production Function Parameters and Productivity

Here, I show how to recover the rest of the parameters of the production function, including productivity and its Markov process. We can do it in several ways, depending on our object of interest.

In the literature, it is not uncommon to assume that productivity follows a Markov process. That is,

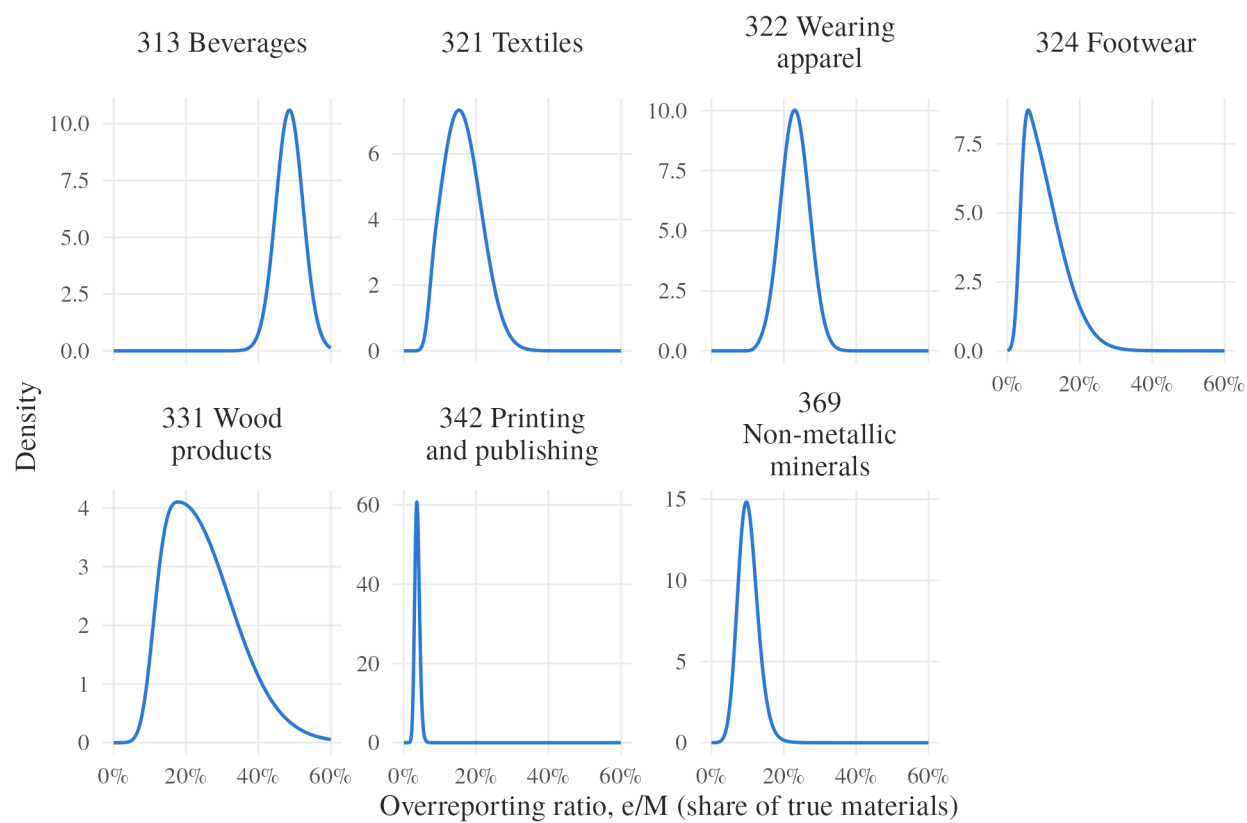


Figure 1: Deconvolved density of the overreporting ratio e/M by industry, unincorporated firms.

$$\omega_{it} = g(\omega_{it-1}) + \eta_{it} \quad (12)$$

Let $\ln X_{it} = x_{it}$. With Equation 9, we can form the following observed variable,

$$\begin{aligned} \mathcal{W}_{it} &\equiv y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) \\ &= y_{it} - \beta(m_{it} + u_{it} - u_{it} + \varepsilon_{it}) \\ &= y_{it} - \beta m_{it} - \beta \varepsilon_{it} \\ &= \alpha_K k_{it} + \alpha_L l_{it} + \omega_{it} + (1 - \beta) \varepsilon_{it} \end{aligned} \quad (13)$$

Here, intuitively, we are trading unobservables, the unobserved tax evasion for the measurement error. Also note that it is not necessary to learn tax evasion to recover productivity. This is useful if a practitioner is only interested in productivity and not in tax evasion.

By replacing ω_{it} from Equation 13 in Equation 12, we get

$$\mathcal{W}_{it} = (1 - \beta) \varepsilon_{it} + \alpha_K k_{it} + \alpha_L l_{it} + g(\mathcal{W}_{it-1} - (1 - \beta) \varepsilon_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) + \eta_{it} \quad (14)$$

If $g(\xi) = \gamma_0 + \gamma_1 \xi$, to learn α and g , we need to find instruments Z such that,

$$\mathbb{E}[\eta_{it} + (1 - \beta) \varepsilon_{it} + \gamma_1 (1 - \beta) \varepsilon_{it-1} | Z] = 0. \quad (15)$$

Table 7 displays candidates for Z .

Table 7: Orthogonality by residuals

	Orthogonality
η_{it}	$k_{it}, l_{it}, \varepsilon_{it}, k_{it-1}, l_{it-1}, \varepsilon_{it-1}, \omega_{it-1}, \mathcal{W}_{it-1}, m_{it-1}, m_{it-1}^*$
ε_{it}	$k_{it}, l_{it}, \omega_{it}, m_{it}, m_{it}^*, \eta_{it}, k_{it-1}, l_{it-1}, \varepsilon_{it-1}, \omega_{it-1}, \mathcal{W}_{it-1}, m_{it-1}, m_{it-1}^*$
ε_{it-1}	$k_{it}, l_{it}, \omega_{it}, \varepsilon_{it}, m_{it}, m_{it}^*, \eta_{it}, k_{it-1}, l_{it-1}, \omega_{it-1}, \mathcal{W}_{it}, m_{it-1}, m_{it-1}^*$

Once the parameters of the production function are identified, we can use the following observed variable to recover ω_{it} ,

$$\begin{aligned} \widetilde{\mathcal{W}}_{it} &\equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} \\ &= \omega_{it} + (1 - \beta) \varepsilon_{it} \end{aligned} \quad (16)$$

4.1 Production function estimates

First, because tax evasion artificially inflates the intermediates' cost share of revenues, lower output elasticities of intermediates than traditional methods are expected. Table 8 shows that correcting for tax evasion results in smaller output elasticities of intermediates than both OLS and GNR in all seven industries.

The elasticities of capital and labour vary across industries, with some being lower, some higher, and others quite similar. This is expected. The direction of the bias in the elasticities of capital and labour depends on whether these inputs are complements or substitutes of intermediate inputs. This relationship will vary by industry. Put differently, for different industries, the cross-elasticity between intermediates and capital, and intermediates and labour will be different in direction and magnitude. Since these cross-elasticities are not included in the CD production function, the elasticities of capital and labour should reflect different degrees of these variations once intermediates are corrected for overreporting.

Productivity estimates in this section use the $\tilde{\mathcal{W}}_{it-2}$ estimates of Table 8.

Correcting for overreporting also reduces dispersion in productivity. The 90th-to-10th-percentile ratio of $\exp(\omega_{it})$ is 1.70 to 2.30 with the correction, against 2.38 to 6.89 for GNR, about 40 percent lower on average (range 18 to 75 percent across the seven industries; Table 9).

Correcting for overreporting also makes productivity more persistent. With instrument $\tilde{\mathcal{W}}_{it-2}$, the AR(1) coefficient of ω_{it} is 0.90 to 0.96 across the seven industries, against 0.81 to 0.89 for GNR, so the half-life of a productivity shock, $\ln 0.5 / \ln \hat{\gamma}_1$, is 7 to 15 years instead of 3 to 6 years: between 1.6 and 3.1 times longer, depending on the industry (Table 9).

Three caveats apply to Table 9. First, the P-spline penalty smooths the deconvolved density, which can understate its dispersion, whereas the firm-level GNR estimates of ω_{it} carry estimation noise, which can overstate it. Second, Gandhi et al. (2020) define productivity in levels as $\exp(\omega_{it} + \varepsilon_{it})$, with ε_{it} an ex-post output shock. Here, ε_{it} is measurement error in output, so productivity is $\exp(\omega_{it})$. The GNR estimates in Table 9 use the same convention, so both compare ω_{it} with ω_{it} , and their published productivity ratios are not directly comparable with these. Third, the two persistence estimates come from different estimators: the corrected $\hat{\gamma}_1$ is an instrumental-variables estimate within the production-function step, whereas the GNR $\hat{\gamma}_1$ is a least-squares regression of firm-level ω_{it} on its lag, which estimation noise in ω_{it} biases towards zero. Part of the gap in persistence, and so in half-lives, may reflect this difference rather than overreporting.

5 Fiscal Policy Evaluation: The 1983 Reform

Firms are known to overreport costs when tax enforcement tightens, which raises the expected cost of evading, but less is known about the benefit side: whether they overreport more when the tax rate,

Table 8: Production function estimates by method: evasion-corrected with instrument $\tilde{\mathcal{W}}_{it-2}$, with m_{it-1}^* , and with both (joint efficient GMM), and uncorrected GNR and OLS. Below each corrected estimate, the 95 percent test-inversion regions (sharp and conservative).

Industry	$\hat{\beta}$	$\tilde{\mathcal{W}}_{it-2}$		m_{it-1}^*		Joint efficient GMM		GNR			OLS		
		$\hat{\alpha}_K$	$\hat{\alpha}_L$	$\hat{\alpha}_K$	$\hat{\alpha}_L$	$\hat{\alpha}_K$	$\hat{\alpha}_L$	$\hat{\beta}$	$\hat{\alpha}_K$	$\hat{\alpha}_L$	$\hat{\beta}$	$\hat{\alpha}_K$	$\hat{\alpha}_L$
313	0.287	0.406	0.307	0.531	0.145	0.302	0.166	0.320	0.337	0.296	0.739	0.121	0.267
sharp	[0.269, 0.306]	[0.295, 0.575] (·, 0.520]		[0.195, ·) (·, 0.430]		empty empty							
cons.	[0.257, 0.321]	[0.095, 0.660] (·, 0.665]		(·, ·) (·, ·)		[0.100, 0.385] [0.015, 0.240]							
321	0.385	0.300	0.349	0.311	0.290	0.297	0.328	0.435	0.225	0.340	0.681	0.093	0.274
sharp	[0.357, 0.416]	[0.210, 0.375] [0.230, 0.485]		[0.210, 0.470] [0.145, 0.420]		[0.230, 0.355] [0.240, 0.425]							
cons.	[0.338, 0.440]	[0.185, 0.390] [0.200, 0.525]		[0.175, ·) [0.090, 0.500]		[0.180, 0.375] [0.205, 0.460]							
322	0.335	0.182	0.392	0.167	0.394	0.161	0.373	0.417	0.175	0.343	0.578	0.047	0.397
sharp	[0.288, 0.391]	[0.085, 0.265] [0.320, 0.470]		[0.090, 0.240] [0.325, 0.465]		[0.075, 0.235] [0.315, 0.430]							
cons.	[0.258, 0.435]	[0.055, 0.285] [0.305, 0.490]		[0.070, ·) [0.310, 0.575]		[0.045, 0.260] [0.295, 0.450]							
324	0.412	0.119	0.427	0.710	0.470	0.123	0.433	0.451	0.131	0.406	0.705	0.013	0.300
sharp	[0.365, 0.465]	(·, 0.270] [0.190, 0.570]		[0.010, ·) [0.235, 0.655]		(·, 0.290] [0.140, 0.570]							
cons.	[0.335, 0.506]	(·, 0.305] [0.095, 0.605]		(·, ·) [0.120, 0.715]		(·, 0.320] [0.065, 0.600]							
331	0.324	0.076	0.554	0.112	0.584	0.071	0.544	0.404	0.102	0.453	0.561	0.064	0.429
sharp	[0.263, 0.398]	(·, 0.200] [0.400, 0.700]		(·, ·) [0.450, ·)		(·, 0.220] [0.390, 0.690]							
cons.	[0.228, 0.459]	(·, 0.240] [0.350, 0.740]		(·, ·) [0.410, ·)		(·, 0.255] [0.350, 0.720]							
342	0.333	0.292	0.408	0.266	0.401	0.265	0.402	0.355	0.221	0.476	0.578	0.066	0.430
sharp	[0.294, 0.377]	[0.215, 0.360] [0.305, 0.530]		(·, ·) [0.150, 0.955]		[0.210, 0.310] [0.335, 0.480]							
cons.	[0.269, 0.412]	[0.190, 0.375] [0.280, 0.570]		(·, ·) [0.065, ·)		[0.180, 0.335] [0.305, 0.525]							
369	0.244	0.333	0.549	0.327	0.639	0.354	0.326	0.279	0.254	0.557	0.495	0.128	0.519
sharp	[0.207, 0.288]	[0.210, 0.485] [0.210, 0.740]		[0.165, ·) [0.330, ·)		(·, 0.450] [0.045, 0.535]							
cons.	[0.184, 0.324]	(·, 0.545] (·, 0.785]		(·, ·) [0.185, ·)		(·, 0.530] (·, 0.630]							

Evasion-corrected columns use the first stage on the log materials share net of sales taxes; $\hat{\beta}$ is the corporations' first-stage estimate, common to the three evasion-corrected columns; below it, its 95% test-inversion regions from the corporations' share moment (the production-function moments are exactly identified given β and profiled): sharp (χ_1^2) and conservative (χ_2^2). Point estimates: the minimum of the test-inversion statistic, with the covariance of the moments re-estimated at each candidate (α_K, α_L), the statistic whose regions are shown. $\tilde{\mathcal{W}}_{it-2}$ (the headline) and m_{it-1}^* : one instrument each (exactly identified, statistic zero at the estimate); below them, the projections of the 95% test-inversion regions for (α_K, α_L) (0.005 grid over $[0, 1]^2$): sharp (χ_2^2 , credit for profiling γ_0, γ_1) and conservative (χ_4^2). Joint efficient GMM: both instruments together; sharp (χ_3^2) and conservative (χ_2^2) regions. Industry 313's joint sharp region is empty (statistic at the estimate $9.46 > 7.81$). GNR and OLS: uncorrected estimates on the same sample (GNR's share regression on the log materials share net of sales taxes). (· or ·): the bound of the parameter space (0 or 1) is not rejected.

Table 9: Productivity corrected for tax evasion (deconvolved, instrument $\tilde{\mathcal{W}}_{it-2}$) and uncorrected (GNR), by industry.

Industry	Method	Mean	SD	Median	75/25	80/20	90/10	95/5	Skewness	$\hat{\gamma}_1$
313	Corrected	27.8	10.9	26.0	1.36	1.49	2.13	3.12	0.21	0.90
	GNR	43.7	22.6	39.0	1.57	1.80	2.99	4.44	-0.02	0.85
321	Corrected	16.5	6.7	15.2	1.41	1.56	2.30	3.39	-0.04	0.95
	GNR	21.2	11.0	19.0	1.82	2.15	3.38	5.31	-0.34	0.87
322	Corrected	57.8	15.9	57.6	1.42	1.58	2.16	2.73	-0.48	0.94
	GNR	34.9	16.5	33.3	1.80	2.13	3.59	5.65	-0.70	0.88
324	Corrected	42.4	11.9	40.8	1.32	1.44	1.94	2.49	0.28	0.96
	GNR	28.8	10.4	27.5	1.53	1.74	2.38	3.08	-0.17	0.87
331	Corrected	85.5	22.6	83.6	1.28	1.38	1.76	2.35	-0.27	0.91
	GNR	48.9	24.5	44.1	1.89	2.25	3.72	5.68	-0.50	0.84
342	Corrected	19.1	4.0	19.0	1.25	1.33	1.70	2.13	-0.30	0.93
	GNR	24.2	11.5	22.2	1.69	1.92	2.88	4.14	-0.20	0.81
369	Corrected	19.0	5.6	18.6	1.30	1.39	1.75	2.41	-0.65	0.93
	GNR	32.8	22.9	28.3	2.69	3.33	6.86	10.95	-0.49	0.89

Productivity in levels, $\exp(\omega_{it})$; percentile ratios as in Gandhi, Navarro and Rivers (2020, Table 3); skewness of ω_{it} ; $\hat{\gamma}_1$: persistence, the AR(1) coefficient of ω_{it} (corrected: IV estimate within the production function step; GNR: OLS on firm-level ω_{it}). Corrected: moments of the deconvolved density of ω (penalized B-spline deconvolution of $\tilde{\mathcal{W}}_{it} = \omega_{it} + (1 - \beta) \varepsilon_{it}$), with the production function estimates of instrument $\tilde{\mathcal{W}}_{it-2}$. GNR: firm-level ω_{it} from the uncorrected GNR (2020) Cobb-Douglas estimation, with the measurement error removed. Both exclude ε . Log materials share net of sales taxes, juridical organization codes 6–9 excluded. Industries: the seven selected in the deconvolution chapter.

which sets how much each peso of fictitious inputs saves, goes up. I investigate this question and find evidence that input overreporting rises when the tax rate increases. In 1983, Colombia raised the sales-tax rate in most manufacturing industries while others remained exempt. Overreporting by unincorporated firms rose after the reform, reaching about 8 percent of true materials by 1987, and the increase was concentrated in the industries where the rate went up.

Whether firms evade more when tax rates rise matters for policymakers. If evasion rises with the rate, a tax increase raises less revenue than its mechanical effect suggests, and evasion affects the marginal efficiency cost of raising tax rates (Slemrod 2019, 907). The concern is especially relevant for sales and value-added taxes, whose credit on purchases is a natural target of fictitious deductions (Carrillo et al. 2022). Yet recent empirical work says more about enforcement than about tax rates.

The 1983 reform changed statutory rates rather than enforcement, which makes it a rare setting in which to study this question. Governments have agreed to randomize letters, audits and information reporting, but not tax rates (Slemrod 2019), and among the studies Slemrod reviews, the evidence on tax rates comes mostly from misreported imports (Fisman and Wei 2004; Handley and Moore 2017). The same literature shows that firms adjust reported costs: when tax authorities pushed firms to report more revenue, Ecuadorian firms raised reported costs by 96 and Armenian firms 70–80 cents per dollar of additional revenue (Carrillo et al. 2017; Asatryan and Peichl 2017).

In the model, the sales tax rate shifts the marginal benefit of overreporting materials: a higher rate on purchases, τ_P , raises the credit claimed on each peso of fictitious inputs (Section 1). Although the model has no income tax, the same logic extends to it: overreported inputs also lower reported profits and, thus, the income-tax base. An increase in either rate therefore raises the incentive to overreport inputs, and a decrease lowers it.

The 1983 fiscal policy changed incentives *across* industries and *within* industries. Across industries, the sales-tax (ST) rate increased from 6 to 10 percent for most ST-liable industries, while ST-exempt industries, food products, remained exempt (Section 2.4). Within industries, the income tax changed by juridical organization (JO), and in the opposite direction. *Ley 9 de 1983* cut individual income-tax rates by 4.6 percentage points on average across income brackets relative to the 1982 schedule, and thus proprietorships faced lower rates (Perry and Cárdenas 1986). LLCs received a smaller cut, from 20 to 18 percent (Perry and Cárdenas 1986). Overreporting inputs lowers both taxes, through a larger credit on purchases and a smaller income-tax base, so the two changes pull the incentive to overreport in opposite directions. In ST-liable industries, the net incentive is therefore ambiguous for both: proprietorships were compensated for much of the sales-tax increase through the income tax, while LLCs were insufficiently compensated. In ST-exempt industries, the income-tax cut alone lowered the incentive for proprietorships and changed it little for LLCs, whose cut was small.

Because the income-tax cut differed between proprietorships and LLCs, the reform's incentives also varied along the size distribution. Proprietorships have a single owner and are the smallest. Tax authorities find sole proprietorships and small businesses hard to penetrate, and evasion in

Table 10: Average individual income-tax rates by taxable income, 1982 schedule (Decreto 2809) vs. the 1983 schedule that stayed (Decreto 397, Ley 9 de 1983), transcribed from Perry and Cárdenas (1986), Cuadro III.1.

Taxable income (thousand 1982 pesos)	1982 schedule	1983 schedule	Change (pp)	Change (%)
200	5.14%	0.24%	-4.90	-95.3
300	10.03%	6.92%	-3.11	-31.0
400	15.65%	10.14%	-5.51	-35.2
500	17.45%	12.80%	-4.65	-26.6
600	21.01%	15.83%	-5.18	-24.7
800	26.28%	21.03%	-5.25	-20.0
1,000	30.06%	24.85%	-5.21	-17.3
1,500	35.61%	31.46%	-4.15	-11.7
2,000	39.03%	35.30%	-3.73	-9.6
Average			-4.63	-22.0^a

Average income-tax rates for an individual declarant with three dependants. 1982: Decreto 2809; 1983: Decreto 397 (Ley 9 de 1983).

Source: Perry and Cárdenas (1986), vol. 1, Cuadro III.1.

^a Excluding the lowest bracket, whose rate falls to almost zero (-95.3%).

Table 11: Implied marginal income-tax rates by income interval, 1982 vs. 1983 schedules, computed from the average rates in Table 10 as the change in tax over the change in income between consecutive points. Each figure is an average over the interval.

Taxable income interval (thousand 1982 pesos)	1982 schedule	1983 schedule	Change (pp)
200–300	19.8%	20.3%	+0.5
300–400	32.5%	19.8%	-12.7
400–500	24.6%	23.4%	-1.2
500–600	38.8%	31.0%	-7.8
600–800	42.1%	36.6%	-5.5
800–1,000	45.2%	40.1%	-5.0
1,000–1,500	46.7%	44.7%	-2.0
1,500–2,000	49.3%	46.8%	-2.5

Implied marginal rates: the change in tax over the change in income between consecutive points of Perry and Cárdenas (1986), Cuadro III.1 (average rates). Each figure is an average over the interval.

them is more sustainable (Slemrod 2019, 940). At the median, LLCs have about two and a half times proprietorships' labour and three times their gross output. LLCs have several owners and non-tradable shares. In Spain, firms bunch just below the revenue threshold above which the Large Taxpayers' Unit monitors them more closely (Almunia and Lopez-Rodriguez 2018; Slemrod 2019, 925–26). At the top, corporations are several times larger than LLCs in capital, labour and gross output, and their shares are tradable. In addition, the government keeps corporations under close supervision and requires them to have an auditor. Lastly, since evidence suggests stock prices of public companies fall when news of their involvement in tax shelters becomes public (Hanlon and Slemrod 2009; Slemrod 2019, 909), corporations might have lower incentives to evade. In this chapter corporations serve as the benchmark of non-evaders, on the assumption that audited statements, third-party reporting and the market's reaction keep them from overreporting.

5.1 Empirical approach

From the materials FOC, for corporations, the truth-reporters, i in industry j at year t , the log share of materials' cost over sales is

$$s_{ijt}^C = \ln \beta_j - \varepsilon_{ijt} \quad (17)$$

where $s_{ijt} \equiv \ln((1 - \tau_{P,ijt})\rho_t M_{ijt}^* / ((1 - \tau_{S,ijt})P_t Y_{ijt}))$ is the log share of materials' cost over sales, both net of sales taxes (Section E), β_j is the output elasticity of materials of industry j , and ε_{ijt} is the error term. For unincorporated firms, the evaders,

$$s_{ijt}^N = \ln \beta_j + u_{ijt} - \varepsilon_{ijt} \quad (18)$$

where $u_{ijt} = \ln(1 + e_{ijt}/M_{ijt})$ is the overreporting of materials' cost due to tax evasion.

Let D_{ijt}^N be a dummy indicating whether the firm is an unincorporated firm. Then,

$$\begin{aligned} (1 - D_{ijt}^N)s_{ijt} + D_{ijt}^N s_{ijt} &= (1 - D_{ijt}^N)(\ln \beta_j - \varepsilon_{ijt}) + D_{ijt}^N (\ln \beta_j + u_{ijt} - \varepsilon_{ijt}) \\ s_{ijt} &= \ln \beta_j + D_{ijt}^N u_{ijt} - \varepsilon_{ijt} \end{aligned} \quad (19)$$

Taking expectations,

$$E[s_{ijt}|D^N = 0, j] = \ln \beta_j, \quad E[s_{ijt}|D^N = 1, j] - E[s_{ijt}|D^N = 0, j] = \mu \quad (20)$$

identifies the average overreporting level, $\mu \equiv E[u_{ijt} | D_{ijt}^N = 1]$, a population expectation; its estimate is $\hat{\mu}$.

In the case of the fiscal policy changes, I am interested in the changes in average tax evasion after the 1983 fiscal reform. Taking the difference of the expectations between year t' and $t = 1983$,

$$\begin{aligned} s_{ijt'} - s_{ij,1983} &= D_{ijt}^N (u_{ijt'} - u_{ij,1983}) - (\varepsilon_{ijt'} - \varepsilon_{ij,1983}) \\ E[s_{ijt'} - s_{ij,1983} | D^N = 1] &= \mu_{t'} - \mu_{1983} \equiv \Delta \mu_{t'} \end{aligned} \quad (21)$$

where μ_t is the annual average tax evasion of year t , and $\Delta \mu_t$ is the change in average tax evasion between year t and 1983, the last full year before the reform took effect (Section 2.4).

The identifying assumptions are explicit in Equation 19 and Equation 21: (i) corporations do not overreport (Equation 17); (ii) firms within an industry share a common Cobb-Douglas technology, with a constant output elasticity of materials β_j (Equation 17 and Equation 18); and (iii) the ex-post shock ε_{ijt} is independent of evasion, true materials and juridical organization, with $E[\varepsilon_{ijt} | D_{ijt}^N, t] = 0$.

Even though Equation 21 resembles a traditional difference-in-differences design, the result does not require parallel trends for identification. Under (i)-(iii), Equation 19 and Equation 21 show that the difference between unincorporated firms and corporations in year t is the average overreporting itself, μ_t , and changes in μ_t over time, including before 1983, are changes in evasion.

To estimate Equation 19, I use the following regression specification,

$$s_{ijt} = \ln \beta_j + \sum_{t=1981}^{1991} \mu_t D_{ijt}^N \mathbb{1}[\text{year} = t] + \zeta_{ijt} \quad (22)$$

where μ_t is the average tax evasion of unincorporated firms in year t ; $\ln \beta_j$ enter as industry fixed effects, estimated jointly with the μ_t , so no first-stage estimate enters the regression and its standard errors need no bootstrap or generated-regressor correction; and $\zeta_{ijt} = v_{ijt} - \varepsilon_{ijt}$, where $u_{ijt} = \mu_t + v_{ijt}$ and v_{ijt} is the idiosyncratic component of tax evasion that is not captured by the year dummies. Likewise, I estimate Equation 21 with the following regression,

$$s_{ijt} = \ln \beta_j + \mu D_{ijt}^N + \sum_{t=1981}^{1991} \mathbb{1}[t \neq 1983] \Delta \mu_t D_{ijt}^N \mathbb{1}[\text{year} = t] + \zeta_{ijt} \quad (23)$$

where $\Delta \mu_t$ is the difference in tax evasion of unincorporated firms between year t and 1983, the last full year before the reform took effect, and μ is the average tax evasion of unincorporated firms in 1983. The outcome is the share net of sales taxes, so the estimates exclude the mechanical effect of the rate change on the share; Section E estimates that effect.

The 1983 fiscal reform changed incentives to evade taxes over time *differently* for firms in ST-liable and ST-exempt industries. Most ST-liable industries faced an ST rate *increase* from 6 to

10 percent, while ST-exempt industries *remained exempt* before and after the reform. Note that firms in *all* industries are subject to the income tax, which also generates incentives to evade taxes by overreporting input costs: lower reported profits due to higher input costs reduce the tax base, and thus the tax liability. Therefore, the income tax and the ST generate tax evasion incentives for firms in ST-liable industries, while only the income tax does for firms in ST-exempt industries. For unincorporated firms,

$$\begin{aligned} s_{ijt}^N &= \ln \beta_j + u_{ijt}^{ST} + u_{ijt}^{IT} - \varepsilon_{ijt} & j \in \text{ST-Liable} \\ s_{ijt}^N &= \ln \beta_j + u_{ijt}^{IT} - \varepsilon_{ijt} & j \in \text{ST-Exempt} \end{aligned} \quad (24)$$

where u_{ijt}^{ST} and u_{ijt}^{IT} stand for the overreporting due to the ST and the income tax (IT), respectively. Let D_{ijt}^E be a dummy indicating whether the industry is ST-exempt and D_{ijt}^L a dummy indicating whether the industry is ST-liable, such that $D_{ijt}^E + D_{ijt}^L = 1$. Combining Equation 17 with Equation 24,

$$s_{ijt} = \ln \beta_j + D_{ijt}^N D_{ijt}^E u_{ijt}^{IT} + D_{ijt}^N D_{ijt}^L (u_{ijt}^{ST} + u_{ijt}^{IT}) - \varepsilon_{ijt} \quad (25)$$

Taking expectations,

$$\begin{aligned} E[s_{ijt}|D^N = 1, D^E = 1, j] - \ln \beta_j &= \mu_E^{IT} \\ E[s_{ijt}|D^N = 1, D^L = 1, j] - \ln \beta_j &= \mu_L^{ST} + \mu_L^{IT} \equiv \mu_L^{ST+IT} \end{aligned}$$

identifies the average overreporting by firms in ST-exempt industries, μ_E^{IT} , and in ST-liable industries, μ_L^{ST+IT} . If I assume $\mu_E^{IT} = \mu_L^{IT}$, I can also identify the overreporting due to ST, μ_L^{ST} . Since income-tax rates depend on the juridical organization and not on the industry, I assume that the income tax generates the same incentives to evade taxes across industries, whether they are exempt of ST or not. Because I aggregate different juridical organizations in exempt and liable industries into unincorporated firms, this assumption might break if the composition of juridical organizations is non-trivially different between exempt and liable industries. The regression versions of Equation 22 and Equation 23 interact the year dummies with $D_{ijt}^N D_{ijt}^E$ and $D_{ijt}^N D_{ijt}^L$, giving one path per industry type, $\mu_{E,t}$ and $\mu_{L,t}$ (and $\Delta \mu_{E,t}$, $\Delta \mu_{L,t}$ relative to 1983). Splitting unincorporated firms further by JO, into LLCs and proprietorships, gives four such paths.

5.1.1 Changes in true materials cannot explain the pattern

Since $u_{ijt} = \ln(1 + e_{ijt}/M_{ijt})$, the empirical exercise remains valid as long as changes in s are not explained by changes in M (or ε), but only by e or e/M : changes in the proxy should be explained by changes to the incentives to adjust overreporting, not by firms adjusting their optimal choice of true inputs.

The alternative argument is that changes in true inputs explain the increase in the overreporting share. Suppose e is unchanged. Then, after the reform, unincorporated firms would have to lower their true inputs, by either an increase in relative prices or productivity shocks. Productivity shocks are unrelated to tax rates, so this is unlikely. According to the FOC, taxes do not show in the optimal choice of firms. Besides, the increase after the reform is gradual, while the reform made only a one-time change. So this alternative argument does not hold. The evasion channel is more direct: an increase in the sales tax rate leads to an increase in the overreporting share because a higher tax rate increases the marginal benefit of overreporting.

5.2 Results: all unincorporated firms

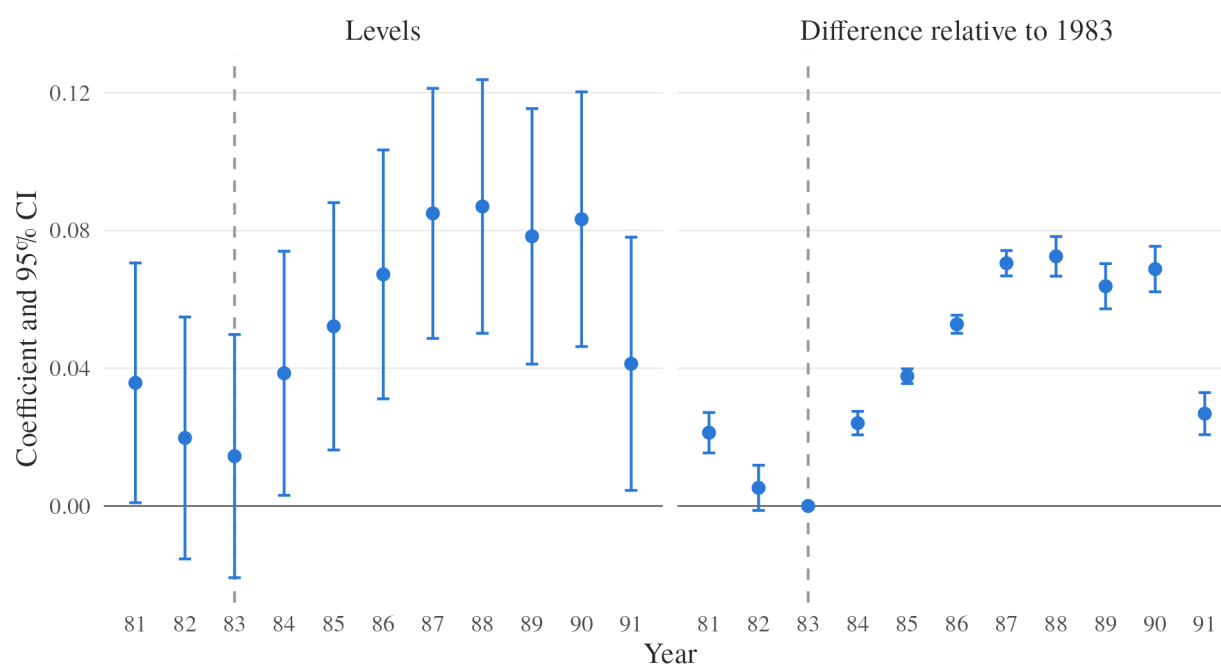


Figure 2: Overreporting of unincorporated firms relative to corporations by year, 1981–1991, all industries. Points are coefficient estimates with 95 percent confidence intervals; standard errors two-way clustered by plant and year; industry fixed effects. Left panel: levels, $\hat{\mu}_t$; right panel: differences with respect to 1983, $\Delta\hat{\mu}_t$. The dashed line marks 1983, the last full year before the reform took effect.

Overreporting rose once the reform raised the benefit of overreporting in most industries: after the sales-tax increase took effect, input overreporting by unincorporated firms increased gradually (Figure 2, Table 12). Input overreporting reached an average of about 9 percent of true materials in 1987, where it stabilized until 1990.

The standard errors of the levels and of the differences over time look very different. In levels, the large standard errors suggest that tax evasion is widely heterogeneous across firms. This

Table 12: Overreporting of unincorporated firms relative to corporations, all industries. Coefficients behind the figure, by year: levels ($\hat{\mu}_t$) and differences with respect to 1983 ($\Delta\hat{\mu}_t$).

Year	Level	Diff. to 1983
1981	0.036* (0.018)	0.021*** (0.003)
1982	0.020 (0.018)	0.005 (0.003)
1983	0.014 (0.018)	–
1984	0.039* (0.018)	0.024*** (0.002)
1985	0.052** (0.018)	0.038*** (0.001)
1986	0.067*** (0.018)	0.053*** (0.001)
1987	0.085*** (0.019)	0.070*** (0.002)
1988	0.087*** (0.019)	0.072*** (0.003)
1989	0.078*** (0.019)	0.064*** (0.003)
1990	0.083*** (0.019)	0.069*** (0.003)
1991	0.041* (0.019)	0.027*** (0.003)

Coefficients with standard errors in parentheses, two-way clustered by plant and year. Industry fixed effects in all columns. Level: $\hat{\mu}_t$; Diff.: $\Delta\hat{\mu}_t = \hat{\mu}_t - \hat{\mu}_{1983}$ (1983 is the reference, –). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Observations: 39,495.

heterogeneity may come from differences in the tax rate due to firm characteristics (industry, JO, and exports), and in the cost of evasion across firms. The differences over time, however, are tight for a mechanical reason. Every year's level $\hat{\mu}_t$ is estimated against the same baseline, so estimation error in that baseline enters every year with the same sign, and the level estimates are strongly positively correlated. Since $\text{Var}(\hat{\mu}_t - \hat{\mu}_{1983}) = \text{Var}(\hat{\mu}_t) + \text{Var}(\hat{\mu}_{1983}) - 2\text{Cov}(\hat{\mu}_t, \hat{\mu}_{1983})$, this shared error cancels in the differences. The correlation affects inference only, not the consistency of the estimates: the regressors are group-by-year dummies, and $E[\zeta_{ijt} \mid D_{ijt}^N, \text{year}] = 0$ because ε is independent of (e, M) and v_{ijt} is, by construction, the deviation from the group-year mean. Standard errors are two-way clustered by plant and year.

5.2.1 Overreporting before the reform

Corporations are a clean comparison group that will not react to the policy changes, because the assumption is that these firms do not evade taxes by overreporting. Thus, I can interpret the estimates as the change in overreporting relative to 1983, the last full year before the reform took effect.

The policy changes of the time were the result of a public debate following the 1981 report of the Bird–Wiesner Mission, a technical mission on Colombia's intergovernmental finances led by Richard Bird (international tax expert) and Eduardo Wiesner (Colombian economist) that recommended better tax administration to reduce evasion (Junguito and Rincón 2004; Bolaños Bolaños 2019).⁸ The reform was enacted at the end of December 1983 and took effect on 1 April 1984 (Section 2.4). These

⁸The mission also recommended stronger collection tools, wider withholding, and more auditing (Bolaños Bolaños 2019, 723).

changes aimed to reduce the fiscal deficit without resorting to money financing and to reactivate investment through tax incentives, while broadening the tax base and simplifying collection (Sánchez Torres and Gutiérrez Sourdis 1994).

A plausible scenario is that firms could have reduced their overreporting during the public debate as a way to signal compliance and avoid further tax increases, and then reverted their behaviour after the reform was enacted. If this is the case, then part of the increase in overreporting after 1983 is a return to previous levels of tax evasion. More data points before 1981 would help to clarify this issue.

5.3 Results: ST-liable vs. ST-exempt industries

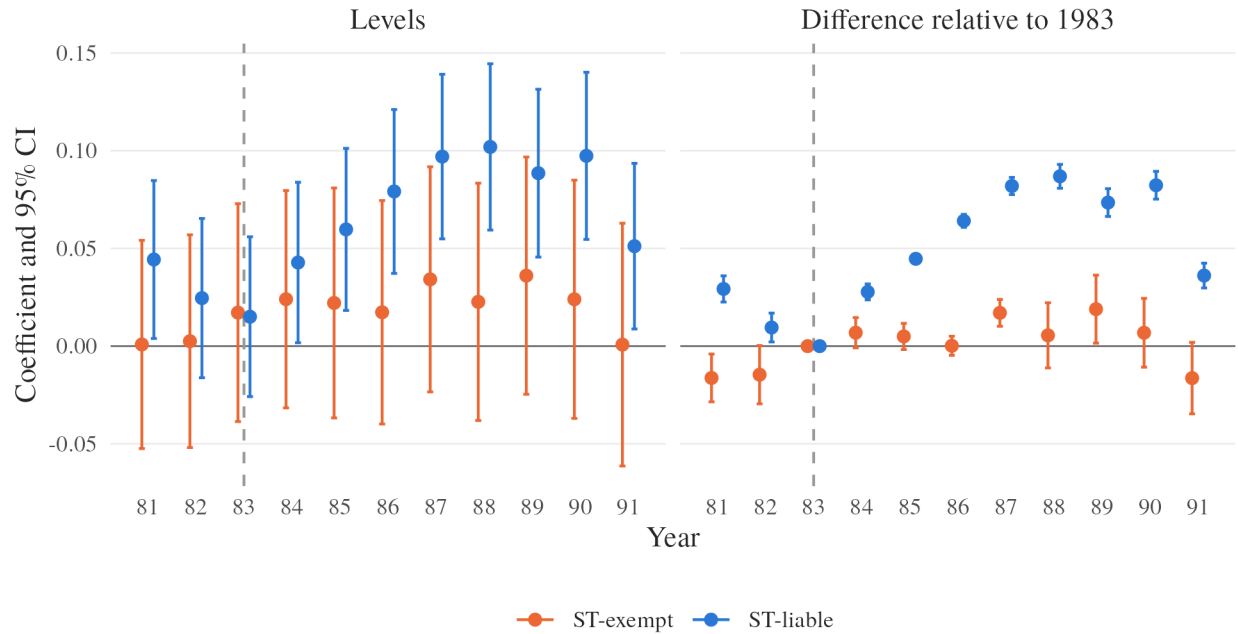


Figure 3: Overreporting of unincorporated firms relative to corporations by year, 1981–1991, in ST-exempt and ST-liable industries. Points are coefficient estimates with 95 percent confidence intervals; standard errors two-way clustered by plant and year; industry fixed effects. Left panel: levels, $\hat{\mu}_t$; right panel: differences with respect to 1983, $\Delta \hat{\mu}_t$. The dashed line marks 1983, the last full year before the reform took effect.

The increase is concentrated where the sales tax, and with it the benefit of overreporting, went up. After the fiscal reform of 1983, unincorporated firms in ST-liable industries increased tax evasion significantly, while those in ST-exempt industries increased it only slightly (Figure 3, Table 13). In ST-liable industries, overreporting rose gradually until 1987 and stabilized until 1990. In ST-exempt industries, the increase was small, at most two percentage points (1987 and 1989). The increase is *larger* for ST-liable firms, the most affected by the reform.

Table 13: Overreporting of unincorporated firms in ST-exempt and ST-liable industries. Coefficients behind the figure, by year: levels ($\hat{\mu}_t$) and differences with respect to 1983 ($\Delta\hat{\mu}_t$).

Year	ST-exempt		ST-liable	
	Level	Diff.	Level	Diff.
1981	0.001 (0.027)	-0.016** (0.006)	0.044* (0.021)	0.029*** (0.003)
1982	0.003 (0.028)	-0.015* (0.008)	0.025 (0.021)	0.010** (0.004)
1983	0.017 (0.028)	—	0.015 (0.021)	—
1984	0.024 (0.028)	0.007 (0.004)	0.043* (0.021)	0.028*** (0.002)
1985	0.022 (0.030)	0.005 (0.003)	0.060** (0.021)	0.045*** (0.001)
1986	0.017 (0.029)	0.000 (0.002)	0.079*** (0.021)	0.064*** (0.002)
1987	0.034 (0.029)	0.017*** (0.004)	0.097*** (0.021)	0.082*** (0.002)
1988	0.023 (0.031)	0.006 (0.009)	0.102*** (0.022)	0.087*** (0.003)
1989	0.036 (0.031)	0.019* (0.009)	0.089*** (0.022)	0.073*** (0.004)
1990	0.024 (0.031)	0.007 (0.009)	0.097*** (0.022)	0.082*** (0.004)
1991	0.001 (0.032)	-0.016 (0.009)	0.051** (0.022)	0.036*** (0.003)

Coefficients with standard errors in parentheses, two-way clustered by plant and year. Industry fixed effects in all columns. Level: $\hat{\mu}_t$; Diff.: $\Delta\hat{\mu}_t = \hat{\mu}_t - \hat{\mu}_{1983}$ (1983 is the reference, —). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Observations: 39,495.

This comparison isolates the sales tax as a shifter of the marginal benefit: ST-liable industries faced the rate increase, while ST-exempt industries did not. Firms in ST-liable industries faced a higher sales tax, only partly offset by lower income-tax rates, while firms in ST-exempt industries faced the income-tax cut alone. The larger response in ST-liable industries is consistent with the sales-tax increase dominating. The slight increase in ST-exempt industries, driven by proprietorships, is unexpected; Section 5.4 discusses two possible explanations.

5.4 Results: income-tax incentives, LLCs vs. proprietorships

Within industries, the income-tax cut lowered the benefit of overreporting, by more for proprietorships than for LLCs. LLCs in ST-liable industries show a significant increase in tax evasion, while LLCs in exempt industries show changes of about one percentage point or less (Figure 4, Table 14). LLCs in exempt industries mainly faced a decrease in their entity-level income tax. This decrease would lead to lower incentives to evade taxes. However, since the reduction is small, it is not unreasonable to find only small changes in tax evasion for these industries. In ST-liable industries, LLCs faced two opposing effects: the increase in sales tax rates, which would increase incentives to evade taxes, and the income-tax cut, which would decrease incentives to evade taxes. The evidence suggests that the increase in sales tax rates dominates the effect of the income-tax cut, leading to a significant increase in tax evasion for LLCs in ST-liable industries.

Proprietorships show a more heterogeneous response (Figure 5, Table 15). In ST-liable industries, their overreporting barely changed in 1984–1986, consistent with the income-tax cut compensating

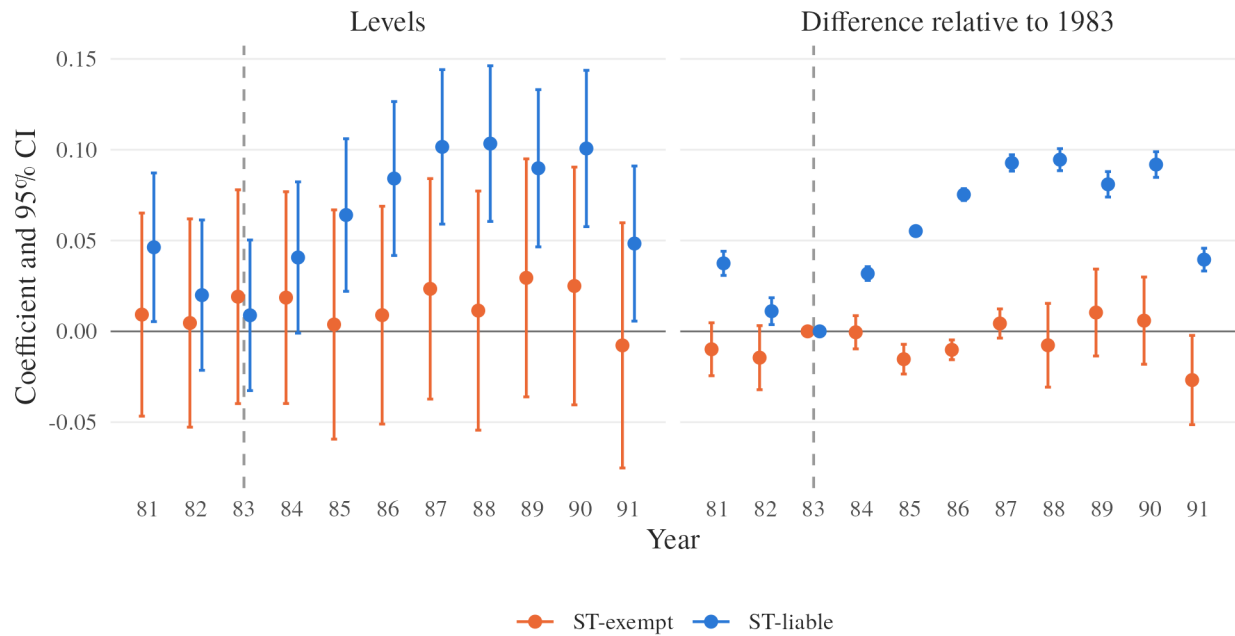


Figure 4: Overreporting of LLCs relative to corporations by year, 1981–1991, in ST-exempt and ST-liable industries. Points are coefficient estimates with 95 percent confidence intervals; standard errors two-way clustered by plant and year; industry fixed effects. Left panel: levels, $\hat{\mu}_t$; right panel: differences with respect to 1983, $\Delta\hat{\mu}_t$. The dashed line marks 1983, the last full year before the reform took effect.

Table 14: Overreporting of LLCs in ST-exempt and ST-liable industries. Coefficients behind the figure, by year: levels ($\hat{\mu}_t$) and differences with respect to 1983 ($\Delta\hat{\mu}_t$).

Year	ST-exempt		ST-liable	
	Level	Diff.	Level	Diff.
1981	0.009 (0.029)	-0.010 (0.007)	0.046* (0.021)	0.037*** (0.003)
1982	0.005 (0.029)	-0.014 (0.009)	0.020 (0.021)	0.011** (0.004)
1983	0.019 (0.030)	—	0.009 (0.021)	—
1984	0.019 (0.030)	-0.001 (0.005)	0.041* (0.021)	0.032*** (0.002)
1985	0.004 (0.032)	-0.015*** (0.004)	0.064** (0.021)	0.055*** (0.001)
1986	0.009 (0.031)	-0.010*** (0.003)	0.084*** (0.022)	0.075*** (0.002)
1987	0.023 (0.031)	0.004 (0.004)	0.102*** (0.022)	0.093*** (0.002)
1988	0.011 (0.034)	-0.008 (0.012)	0.103*** (0.022)	0.095*** (0.003)
1989	0.029 (0.033)	0.010 (0.012)	0.090*** (0.022)	0.081*** (0.004)
1990	0.025 (0.033)	0.006 (0.012)	0.101*** (0.022)	0.092*** (0.004)
1991	-0.008 (0.034)	-0.027* (0.013)	0.048* (0.022)	0.039*** (0.003)

Coefficients with standard errors in parentheses, two-way clustered by plant and year. Industry fixed effects in all columns. Level: $\hat{\mu}_t$; Diff.: $\Delta\hat{\mu}_t = \hat{\mu}_t - \hat{\mu}_{1983}$ (1983 is the reference, —). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Observations: 39,495.

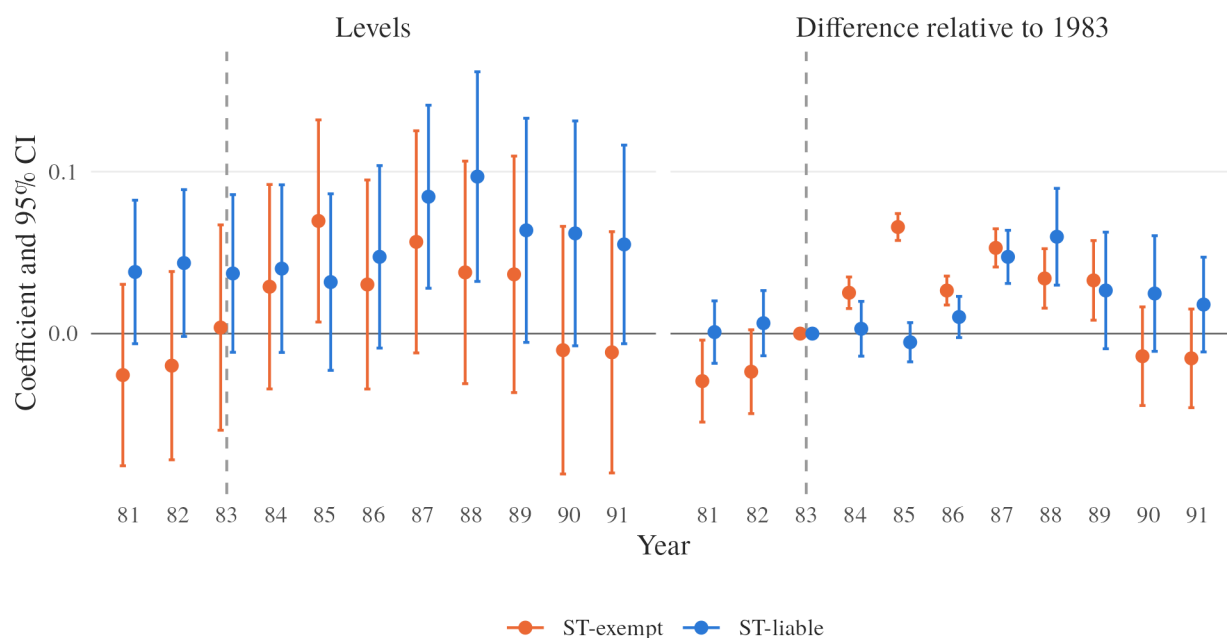


Figure 5: Overreporting of proprietorships relative to corporations by year, 1981–1991, in ST-exempt and ST-liable industries. Points are coefficient estimates with 95 percent confidence intervals; standard errors two-way clustered by plant and year; industry fixed effects. Left panel: levels, $\hat{\mu}_t$; right panel: differences with respect to 1983, $\Delta\hat{\mu}_t$. The dashed line marks 1983, the last full year before the reform took effect.

Table 15: Overreporting of proprietorships in ST-exempt and ST-liable industries. Coefficients behind the figure, by year: levels ($\hat{\mu}_t$) and differences with respect to 1983 ($\Delta\hat{\mu}_t$).

Year	ST-exempt		ST-liable	
	Level	Diff.	Level	Diff.
1981	-0.026 (0.029)	-0.029** (0.013)	0.038 (0.023)	0.001 (0.010)
1982	-0.020 (0.030)	-0.024 (0.013)	0.044* (0.023)	0.006 (0.010)
1983	0.004 (0.032)	—	0.037 (0.025)	—
1984	0.029 (0.032)	0.025*** (0.005)	0.040 (0.026)	0.003 (0.009)
1985	0.070* (0.032)	0.066*** (0.004)	0.032 (0.028)	-0.005 (0.006)
1986	0.030 (0.033)	0.027*** (0.005)	0.047 (0.029)	0.010 (0.006)
1987	0.057 (0.035)	0.053*** (0.006)	0.085** (0.029)	0.047*** (0.008)
1988	0.038 (0.035)	0.034*** (0.009)	0.097** (0.033)	0.060*** (0.015)
1989	0.037 (0.037)	0.033** (0.013)	0.064 (0.035)	0.027 (0.018)
1990	-0.010 (0.039)	-0.014 (0.016)	0.062 (0.035)	0.025 (0.018)
1991	-0.012 (0.038)	-0.015 (0.016)	0.055 (0.031)	0.018 (0.015)

Coefficients with standard errors in parentheses, two-way clustered by plant and year. Industry fixed effects in all columns. Level: $\hat{\mu}_t$; Diff.: $\Delta\hat{\mu}_t = \hat{\mu}_t - \hat{\mu}_{1983}$ (1983 is the reference, —). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Observations: 39,495.

them for much of the sales-tax increase; it rose from 1987, peaking in 1988. In ST-exempt industries, their overreporting rose after 1983, even though they faced only the income-tax cut. This result is unexpected, and I offer two possible explanations. First, the incentive to overreport depends on the marginal tax rate, not the average one, and the 1983 cut in marginal rates was much smaller than the cut in average rates at both ends of the income distribution: close to zero at low incomes and about 2 percentage points at high incomes (Table 11). Second, the 1982–1983 changes stopped individuals from fully deducting their financial expenses (Ocampo and Perry 1983), which gave proprietorships, but not LLCs, a reason to replace a lost deduction with overreported inputs.

Proprietorships' standard errors are also several times those of LLCs, which suggests that they responded less uniformly. This is plausible: proprietorships range from family workshops to young entrepreneurial firms, with varying degrees of process standardization, record-keeping, while LLCs tend to be larger firms with more established processes, and their responses look more homogeneous.

5.5 Takeaways and link to the structural model

Input overreporting rises with the marginal benefit of overreporting. After the 1983 reform raised the sales tax, overreporting by unincorporated firms increased, reaching about 9 percent of true materials by 1987, and the increase was concentrated in ST-liable industries, where the benefit rose. Within those industries it rose most among LLCs, whose small income-tax cut left most of the sales-tax increase in place, while proprietorships, compensated by a larger income-tax cut, barely responded at first. The one result that does not fit, the rise among proprietorships in ST-exempt industries, has two possible explanations in the reform's other provisions (Section 5.4).

This evidence supports the model's key comparative static, $\partial e^* / \partial \tau_P > 0$ (Section 1), which the structural estimation cannot test by regression: overreporting and the tax rate are both built from the same reported materials, so a regression of one on the other is mechanically biased. It also supports the identification of the detection probability in Section 6, which rests on the same source of variation: changes in the sales-tax benefit of overreporting move evasion.

6 The Counterfactual: Claimed Deductions and the Tax Rate

6.1 Partial Equilibrium Claimed-Deductions Curve

How do tax revenues change as the government changes the sales tax rate? As the tax rate increases, the incentives to claim additional fictitious deductions increase, but an increase in evasion raises the probability of detection. Therefore, firms face a trade-off. On the flip side, a decrease in the tax rate reduces the incentives to evade and firms start to decrease the level of overreporting (intensive margin) and some firms start jumping out of evasion altogether (extensive margin). For the government, taking the current observed state as given, what is the marginal change in revenue

as the tax rate changes?

I study this question using the estimates of the structural objects from the firms' tax evasion model. With the parameters at hand, I construct a partial-equilibrium claimed-deductions curve: expected claimed deductions as a function of the tax rate. Claimed deductions are the channel through which evasion moves revenue; every extra peso claimed is a peso of revenue lost.

The curve rises gently on the tax-cut side and steeply on the tax-increase side. A 0.5 percent increase in the rate already raises claimed deductions significantly, by 16–28 percent, lowering revenue, while claimed deductions fall significantly below their current level only after a cut of at least 8 percent.

Given the firm's optimization problem, the government's expected tax revenue is,

$$R_i(\Delta; M_i) = \tilde{\tau}_{it}(P_t Y_{it} - [M_i + (1 - q(e'_i(\Delta)))e'_i(\Delta)]) \quad (26)$$

where $\tilde{\tau}_{it} \equiv (1 + \Delta)\tau_{it}$. The first term in Equation 26 is the revenue collected from sales taxes on sales, $\tilde{\tau}_{it}P_t Y_{it}$. The second term is the credits claimed on sales taxes on purchases. The term includes the firms' true inputs, M_i , plus the counterfactual undetected evasion, $(1 - q(e'_i(\Delta)))e'_i(\Delta)$, at tax rate percentage change Δ . The firm's FOC shows τ directly modifies the marginal benefit of evasion leading to changes in the optimal evasion $e'_i(\Delta)$.

In general, governments can also choose the level of enforcement effort, and therefore they can affect $q(\cdot)$. In addition, governments might collect additional income by penalizing firms caught evading.

Since I lack information on these margins, I hold output and input decisions fixed. The tax on sales then changes only mechanically with the rate, and the behavioural response of revenue runs entirely through claimed deductions, so I focus on

$$-\tilde{\tau}_{it}[M_i + (1 - q(e'_i(\Delta)))e'_i(\Delta)]$$

6.2 Estimating the model

6.2.1 Setup

Using estimates of the production function parameters $(\beta, \alpha_K, \alpha_L)$ and constructed residuals, $\mathcal{V}_{it}$, $\tilde{\mathcal{W}}_{it}$ from Section 3.5 and Section 4, the objective is to recover the parameters of the probability of detection and the cost of evasion function $(q(e), \kappa(e, \omega))$.

As described in Section 1, I follow the classical tax evasion literature building on the seminal work of Allingham and Sandmo (1972). In this literature, the probability of detection, $q(e)$, is increasing in the evasion level. This assumption is consistent with a tax authority that audits a firm whenever its

overreporting exceeds a random threshold.⁹ The cost of evasion, $\kappa_{it}(e, \omega)$, represents the resources of the firm allocated to the evasion endeavour that could have been employed in a productive activity for the firm. The cost of evasion is increasing in the evasion level and convex in productivity. The ψ term, an exogenous cost shock, allows cross-firm heterogeneity.

The cost of evasion, convex in productivity, rationalizes the observed pattern in Carrillo et al. (2022), where evasion is increasing along the size distribution of firms but drops drastically at the very top. In the production function literature, firms are typically assumed to share a common technology and differ only in their productivity, and thus productivity represents a major component of size. In the model, two primitives govern firms' evasion decisions: the probability of detection and the cost of evasion. The probability of detection cannot explain the increase in evasion along the size distribution. Holding the cost of evasion constant, the probability of detection would have to decrease as firms grow in size, meaning bigger firms are less likely to be audited, which does not make economic sense. If anything, the probability of detection could remain flat along the size distribution for all but the very largest firms; higher scrutiny of the very largest firms could add to the drop at the top, but it cannot produce the rise below it.

In contrast, the pattern can be rationalized by the cost of evasion being an economic opportunity cost. Take, for example, the manager's time. For small firms, managers commonly multitask across different productive activities, such as sales, HR, purchasing, and administrative tasks. For these firms, allocating time to tax evasion is costly. As firms increase in size and resources, managers delegate activities according to the firm's division of labour. For large firms, allocating some of the manager's time to evasion is not as costly. However, as firms grow past a certain threshold, departments grow in resources, including the number of employees. Allocating resources to any single activity requires coordination among a larger number of employees in different departments. For example, the records from the warehouse, accounting, purchasing, and finance must match for every fake invoice used to claim deductions. Therefore, the cost of evasion might decrease as firms get bigger but increase past a certain threshold.

Linear detection-probability and evasion-cost functions guarantee a unique interior solution for the evasion level (see Section B). In particular, I assume these functions to be,

$$q(e) = \min\{\lambda e, 1\} \quad (27)$$

$$\kappa_{it}(e_{it}, \omega_{it}) = e_{it} \mathbf{e}^{\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it}} \quad (28)$$

Using the functional forms defined above, the first-order condition with respect to tax evasion for

⁹The probability of detection might also depend on enforcement effort a , $q(e, a)$. However, since I do not observe enforcement, I treat it as constant across firms and time.

evaders (interior solution) becomes,

$$\tau \rho_t (1 - 2\lambda e_{it}) = \mathbf{e}^{\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it}}$$

Let

$$h(e_{it}; \lambda) \equiv \ln(\tau \rho_t) + \ln(1 - 2\lambda e_{it}), \quad e_{it} \in (0, \frac{1}{2\lambda}),$$

Taking logs, the evasion FOC becomes,

$$\psi_{it} = h(e_{it}; \lambda) - \delta_0 + \delta_1 \omega_{it} - \delta_2 \omega_{it}^2 \quad (29)$$

The parameters of interest are $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$.

6.2.2 Estimation

Using the firm-level constructed variables from Section 3.5 and Section 4, I cast the estimation problem in terms of a single unobservable, true materials M_{it} ,

$$e(M) = M_{it}^* - M, \quad \varepsilon(M) = \ln(M_{it}^*/M) - \mathcal{V}_{it}, \quad \omega(M) = \tilde{\mathcal{W}}_{it} - (1 - \beta) \varepsilon(M)$$

Therefore, I rewrite Equation 29 as,

$$\psi(M) = h(e(M)) - \delta_0 + \delta_1 \omega(M) - \delta_2 \omega(M)^2 \quad (30)$$

In addition to depending on a single unobservable, M_{it} , the setup presents two non-standard features. First, it is nonlinear and nonseparable in M through the log term in h , and thus a traditional method of moments cannot help (one cannot interact with an instrument to make the undesired error term go away). Second, the domain of M depends on the parameter being estimated. The support of the latent variable depends on λ because the FOC establishes the ceiling $e < \frac{1}{2\lambda}$ and restricts true materials to $M \in (M_{it}^* - \frac{1}{2\lambda}, M_{it}^*)$.

The method *Entropic Latent Variable Integration via Simulation* (ELVIS) (Schennach 2014) is built for exactly this combination of features.

Let f_Z denote the density of the observed data Z_{it} , and let $f_{M|Z}$ denote a candidate conditional density of M_{it} given Z_{it} , so that $f_{M|Z} f_Z$ is the implied joint density of (M_{it}, Z_{it}) .¹⁰ The question is whether

¹⁰Schennach (2014) states the result for general probability measures. I write it with densities because, in this application, M_{it} is continuous on the known support $(\max\{0, M_{it}^* - \frac{1}{2\lambda}\}, M_{it}^*)$; every candidate conditional distribution therefore has a density there. The reference density $f_{M|Z}^0$ used in estimation is the uniform density on that support: the estimator draws M_{it} uniformly on the support and accepts each draw with probability $\min\{1, \exp[\gamma'(g_{\text{new}} - g_{\text{old}})]\}$;

some choice of $f_{M|Z}$ makes $E_{f_{M|Z}f_Z}[g(M, Z, \theta)] = 0$.

Schennach (2014) shows the abstract *existence* problem of $f_{M|Z}$ is equivalent to a more concrete optimization problem (her Theorem 2.1):

$$\inf_{f_{M|Z}} \|E_{f_{M|Z}f_Z}[g(M, Z, \theta)]\| = 0 \iff \inf_{\gamma \in \mathbb{R}^{d_g}} \|E_{f_Z}[\tilde{g}(Z, \theta, \gamma)]\| = 0 \quad (31)$$

where

$$\tilde{g}(Z, \theta, \gamma) \equiv \frac{\int g(M, Z, \theta) \exp(\gamma' g(M, Z, \theta)) f_{M|Z}^0(M|Z) dM}{\int \exp(\gamma' g(M, Z, \theta)) f_{M|Z}^0(M|Z) dM} \quad (32)$$

Among the conditional distributions of M given Z that satisfy the moment conditions, ELVIS selects the one with maximum entropy relative to an arbitrary reference density $f_{M|Z}^0$. In the optimization problem, $\gamma \in \mathbb{R}^{d_g}$ is the Lagrange multiplier vector on the moment constraints. $f_{M|Z}^0$ only needs to be positive wherever the true conditional density of M is; any choice with the right support gives the same $\hat{\theta}$ (Remark 2.3, Schennach 2014). In this setting, this allows me to avoid specifying f_ψ or f_ε or searching over the space of all conditional distributions of M . Instead, I solve a finite optimization problem.

The true (unobserved) conditional distribution of M satisfies the moment condition at the true θ_0 , so Theorem 2.1 guarantees some γ also solves the right-hand side of Equation 31 at θ_0 . Searching over θ for the value where this finite-dimensional problem has a solution therefore recovers θ_0 , regardless of the chosen $f_{M|Z}^0$, without requiring knowledge of the true conditional distribution of M .

The entropy-maximizing distribution satisfies $E[g(M, Z, \theta_0)] = 0$, but it should not be treated as the actual distribution of M , since many conditional distributions of M given Z satisfy the moment condition (Schennach 2022, 1250). This matters for the counterfactual question. Since expected counterfactual revenue depends on the latent M_{it} , it would be tempting to average it over draws from such a distribution. Instead, I include the counterfactual object as an auxiliary parameter with its own moment condition and estimate it jointly with θ and γ .

During estimation, I use the model's independence assumptions: ε is independent of (M, e, ω) and ψ is independent of (ω, M) , with $E[\varepsilon] = E[\psi] = 0$. They imply the following set of moments,

the chain's average therefore computes $\tilde{g}$ in Equation 32.

$$g(M, Z, \theta) = \begin{bmatrix} \psi(M, Z) \\ \varepsilon(M, Z) \\ \psi(M, Z) \ln M \\ \psi(M, Z) \omega(M, Z) \\ \psi(M, Z) \omega(M, Z)^2 \\ \varepsilon(M, Z) \ln M \\ \varepsilon(M, Z) e(M, Z) \\ \varepsilon(M, Z) \omega(M, Z) \\ \frac{\partial h(e, \lambda)}{\partial \lambda} \varepsilon(M, Z) \end{bmatrix}$$

The nine moments are combined into a single test statistic through the continuously updated estimator (CUE) objective $\hat{L}_n(\theta)$ of Schennach (2014)'s Definition F.1. Her Theorem F.1 shows that, at any θ satisfying the moment condition, $TS(\theta) = 2n\hat{L}_n(\theta)$ is asymptotically bounded above by a $\chi^2_{d_g}$ random variable. Comparing $TS(\theta)$ with the $\chi^2_{d_g}$ critical value therefore gives a conservative test, whose size is at most its nominal level; I call it the conservative test throughout and report as confidence sets the tested points it does not reject. For the nine moments, the 95 percent critical value is $\chi^2_{9,95} = 16.92$. The exact objective and my sign convention relative to hers are in Section A.

The estimation sample has 32,232 unincorporated firm-periods (3,340 at the $\tau = 0$ corner), after dropping the 0.5 percent of non-corner firm-periods with the largest reported materials (Section A).

6.2.3 Estimates

Table 16 displays the estimates of the parameters of interest, using the production-function estimates with instrument m_{it-1}^* from Section 4. The estimation uses all 29 industries, not only the five reported in Section 4; the production-function estimates for every industry, with the same instrument, are in Section C. It also assumes that every unincorporated firm with a positive sales-tax rate on purchases evades. The tests in Section 3 cannot reject the absence of overreporting in some industries, but a failure to reject is inconclusive rather than evidence of compliance, and overreporting rose across all industries pooled after the 1983 reform (Section 5.2).

Summary of findings:

- Probabilities of detection are tiny: the average audit-triggering threshold $1/(2\hat{\lambda}) \approx \text{COP } 920,000$, while median reported materials $\text{Med}[M^*] \approx \text{COP } 8,414$, the 95th-percentile firm COP 128,154, the 99th COP 360,676, and the largest (trimmed) firm COP 701,130.
- The cost function is convex in productivity; both $(\hat{\delta}_1, \hat{\delta}_2)$ display the expected signs.
 - $\hat{\delta}_1 > 0$ and $\hat{\delta}_2 > 0$: the evasion cost function is decreasing in productivity up to $\hat{\omega}^*$, then increasing again
 - $\hat{\omega}^* = 3.98$, the 86th percentile of productivity (forward simulation)

Table 16: Stage-2 estimates of the detection and evasion-cost parameters at the best point, with production-function instrument m_{it-1}^* . TS_{cons} is the conservative test statistic, compared with $\chi^2_{9,.95}$.

	m_{it-1}^* ^a
$\hat{\lambda}$	5.43e-07
$\hat{\delta}_0$	3.464
$\hat{\delta}_1$	4.300
$\hat{\delta}_2$	0.540
$\hat{\omega}^* = \hat{\delta}_1/2\hat{\delta}_2$	3.981
$\max \gamma $	2.644
TS_{cons}	13.860
$\chi^2_{9,.95}$	16.919
Result	Passes

^a Point estimate refined from a joint 3D $(\lambda, \delta_1, \delta_2)$ grid (27 cells).

Table 17: Implied detection probability $q(e) = \hat{\lambda}e$ across the distribution of baseline evasion. The ratio to the median evader does not depend on $\hat{\lambda}$.

	e	$q(e)$	$q(e)/q(\text{Med})$
Mean	1,946	0.11%	4×
Median	495	0.03%	1×
P95	6,912	0.38%	14×
P99	20,099	1.09%	41×
Max	694,645	37.70%	1405×

Implied detection probability $q(e) = \hat{\lambda} \cdot e$ across the evasion distribution, using forward-simulation estimates. Because q is linear in e , $q(e)/q(\text{median}) = e/\text{median}$, which does not depend on $\hat{\lambda}$.

6.2.3.1 What would the estimated detection probabilities imply?

To get a ballpark of what $\hat{\lambda}$ would imply in terms of detection probability, I forward-simulate different percentiles of the evasion distribution and compare their implied detection probabilities. Since $q(e)$ is linear, the ratio of these statistics provides a measure of the relative likelihood of detection. A firm that evades twice as much as another is twice as likely to be caught. Table 17 displays the results.

Key observations:

- Mean $\gg$ Median; a right-skewed evasion distribution
- P99 $\rightarrow$ Max: detection probability jumps from about 1 percent at the 99th percentile to ~37 percent for the most aggressive evader (0.5 percent-trimmed sample)
- A handful of extreme evaders pull $E[e]$ **up** and the fitted $\hat{\lambda}$ **down**

6.2.4 Discussion

- $\hat{\lambda}$ is **very small**: a few aggressive evaders dominate the estimate; the model is trying to fit a straight line over a highly right-skewed distribution.
 - Mechanically, selecting a functional form for the probability of detection imposes a ceiling probability
 - Assuming all unincorporated firms with a non-zero sales tax rate on purchases evade taxes implies that all firms evade below the ceiling (normalization)
- $(\hat{\delta}_1, \hat{\delta}_2)$: seem to be consistently identified by the curvature of the evasion distribution and productivity ω . $\hat{\omega}^*$ is stable across the grid searches in Section A, and $(\hat{\delta}_1, \hat{\delta}_2)$ have the expected signs at every tested point

6.3 Counterfactual design

Let the counterfactual tax rate be $\tilde{\tau}_{it} = (1 + \Delta)\tau_{it}$. Then, given the parameter estimates, the counterfactual evasion is,

$$e'_i(\Delta) = \max\left(0, \frac{\tilde{\tau}_{it}\rho_t - C(\omega_i, \psi_i)}{2\lambda \tilde{\tau}_{it}\rho_t}\right), \quad C(\omega, \psi) \equiv e^{\delta_0 - \delta_1\omega + \delta_2\omega^2 + \psi}$$

Firms might jump out of evasion and optimally select $e'_i = 0$ as Δ adjusts the marginal benefit of evasion. This decision justifies $\max(0, \cdot)$.

Substituting the baseline FOC, e'_i can be written in terms of the firm's baseline evasion e_i ,

$$e'_i(\Delta) = \max\left(0, \frac{1}{1 + \Delta}[e_i + \Delta/(2\lambda)]\right) \quad (33)$$

where $e_i = M_i^* - M_i$ is the firm's baseline evasion at the current state, a function of the latent M_i , and captures the firm heterogeneity defined by (ω, ψ) . Equation 33 shows that the counterfactual evasion is linear in the model's current evasion with slope $1/(1 + \Delta)$ and an intercept, $\Delta/[2\lambda(1 + \Delta)]$, constant across firms.

The counterfactual evasion is a function of the same latent M_i as the main moments system. As briefly mentioned in Section 6.2.2, define firm i 's counterfactual deduction claims as $\text{Claims}_i(\Delta) \equiv \tilde{\tau}_{it}[M_i + (1 - q(e'_i(\Delta)))e'_i(\Delta)]$, the credit claimed on true inputs plus undetected overreporting. I include an auxiliary parameter $\theta_C(\Delta)$, mean deduction claims, with its own moment $E[\text{Claims}_i(\Delta) - \theta_C(\Delta)] = 0$. Claims are measured in nominal COP; each firm-period's claims are deflated by the GDP deflator of its year before averaging, so $\theta_C(\Delta)$ is real mean claims per firm-period. Therefore, the number of moments increases to $d_g = 10$.

To obtain the confidence set of $\theta_C(\Delta)$, I invert the test. At each Δ , θ is fixed at the best point from

Section 6.2.3 and only γ is re-optimized (Aguiar and Kashaev 2020). I grid over candidate θ_C and build the confidence set $\{\theta_C : TS(\Delta, \theta_C) < \chi^2_{10,.95} = 18.31\}$.

6.4 Results

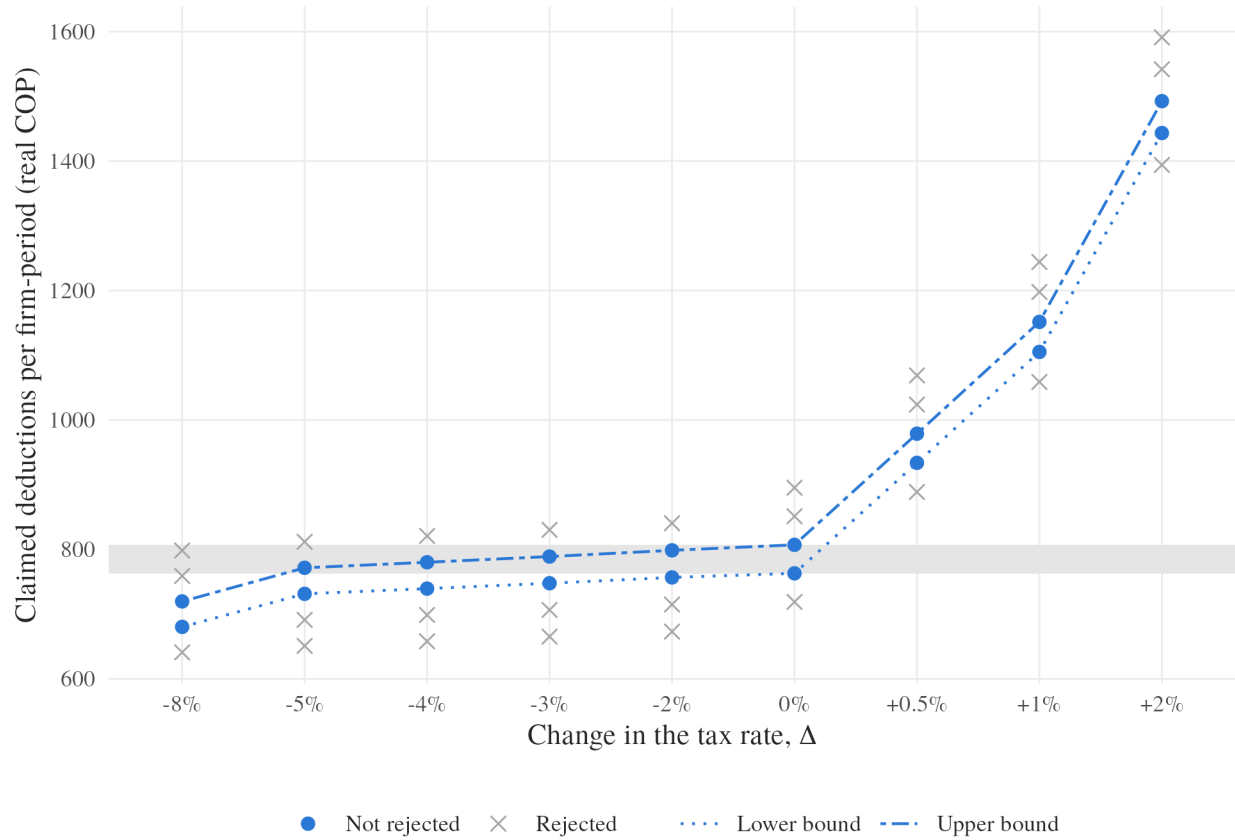


Figure 6: Conservative-test 95 percent confidence sets for real mean deduction claims per firm-period (COP), by tax-rate change Δ . Each marker is a tested candidate value of $\theta_C(\Delta)$: filled circles are not rejected, crosses are rejected. The dotted and dashed lines connect the lowest and highest non-rejected candidates at each Δ . The grey band is the confidence set at current policy ($\Delta = 0$). The structural parameters are fixed at the best point in Table 16; critical value $\chi^2_{10,.95} = 18.31$.

Figure 6 displays the estimated curve of the counterfactual real average claims per firm-period.

Key findings (Table 19):

- A tax increase of 0.5 percent is already statistically distinguishable from current policy, raising claims; a cut lowers claims but only becomes distinguishable at -8 percent.
 - Baseline CI [763, 807]. Cut of 8 percent: [680, 720], change bounded between -15.71 percent and -5.69 percent. Increase of 0.5 percent: [934, 979], change bounded between $+15.70$ percent and $+28.28$ percent.

Table 18: Real mean deduction claims per firm-period (COP), 95 percent conservative-test confidence sets, 9 tested Δ points.

	-8%	-5%	-4%	-3%	-2%	0%	+0.5%	+1%	+2%
Lower	680	731	739	748	757	763	934	1105	1443
Upper	720	772	780	789	799	807	979	1151	1493

Real mean claimed deductions per firm-period (COP): lowest and highest tested values not rejected by the 95% conservative test at each Δ .

Table 19: Change in real mean deduction claims relative to current policy, for the smallest tested tax cut that is distinguishable from baseline and the smallest tested tax increase. The bounds compare the two 95 percent conservative-test confidence sets.

Scenario	Tax change	Claims CI (real COP)	Bounds on the change	Point-estimate change
Baseline (current policy)	0.0%	[763, 807]	–	–
Tax cut (smallest that separates)	-8.0%	[680, 720]	-15.71% to -5.69%	-10.84%
Tax increase (smallest tested)	+0.5%	[934, 979]	+15.70% to +28.28%	+21.27%

Bounds compare the two 95% conservative-test confidence sets.

- Mechanism for the asymmetry: $e'_i(\Delta) = [e_i + \Delta/(2\lambda)]/(1 + \Delta)$ is a horse race between e_i and the constant term $\Delta/(2\lambda)$. With $\hat{\lambda}$ tiny, the constant wins: an increase shifts every firm by about the same large amount; a cut removes evasion firm by firm (small evaders hit zero first), so the aggregate response is gradual.
- This is a property of linear q and linear κ combined with small $\hat{\lambda}$, not of linearity alone. Convex $q = (\lambda e)^k$ and concave $q = 1 - e^{-\lambda e}$ break the uniform-shift result .

6.5 Robustness

- Rejected alternative target: only the undetected-overreporting part of claimed deductions, $\tilde{\tau}_{it}(1 - q(e'_i))e'_i$, leaving out the credit on true inputs. For cuts beyond about 2 percent, most firms stop evading ($e'_i = 0$), so its distribution piles up at zero, and in a coarse grid most tested brackets were inconclusive. Claimed deductions in full remain the target.

6.6 Scope and limits

- Partial equilibrium: K , L , ω , ε and M held fixed; only evasion responds. Large Δ is outside what the exercise can speak to.
- θ fixed at one operating point that is not rejected (conservative test), with production-function instrument m_{it-1}^* .
- $\hat{\lambda}$ is identified mostly from a thin tail of aggressive evaders; the estimate depends on the 0.5

percent trim.

7 Conclusion

Online Appendix

A ELVIS Derivation and Implementation

A.1 Estimation mechanics

- CUE quadratic objective with eigenvalue-truncated weighting matrix.
- **Sales-tax revenue on sales is observed.** The data record the sales tax each firm pays on its sales in each period, which is the observed counterpart of $\tau_{it}P_tY_{it}$ in Equation 26; total revenue in the counterfactual is computed from it directly, held at its baseline value.
- **The λ -score moment enters in bounded form.** The score $h_\lambda(e) \equiv \partial h(e; \lambda) / \partial \lambda = -2e / (1 - 2\lambda e)$ diverges near the FOC ceiling $e \rightarrow 1 / (2\lambda)$, where a single extreme draw can dominate the weighting matrix's eigenvalues and silently collapse the effective moment count. The implementation therefore replaces h_λ with $h_\lambda / (1 - h_\lambda) \in (-1, 0]$ in the ninth moment ($1 - 2\lambda e$ floored at 10^{-6}). Validity is unaffected: any function of e alone is independent of ε under the same argument that licenses the raw score. The transform decays polynomially, not exponentially, so firms stay numerically distinguishable across the whole range of e ; $\exp(h_\lambda)$ and $\tanh(h_\lambda)$ were rejected because both saturate to a constant for most of the sample.
- Per-firm Metropolis chain: $n_{burn} = 1000$, $n_{keep} = 3000$.
- Nelder-Mead optimizer.
- γ is unbounded in the optimizer: Schennach (2014) and Aguiar and Kashaev (2020) both treat $\gamma \rightarrow \pm\infty$ as an expected solution, not a numerical failure.

A.2 Trim cutoff

- Top 0.5 percent of interior (non-corner) firms by M^* dropped before fitting.
- Why: a handful of extreme evaders near the FOC ceiling drive $\hat{\lambda}$ toward zero without it; 0.5 percent is the first cutoff where the 95 percent CI's upper bound stabilizes.

A.3 Search strategy for the joint $(\lambda, \delta_1, \delta_2)$ region

- Search order: λ -grid (profiling δ_1, δ_2), then a (δ_1, δ_2) -grid (profiling λ), then a joint 27-cell $(\lambda, \delta_1, \delta_2)$ cube.

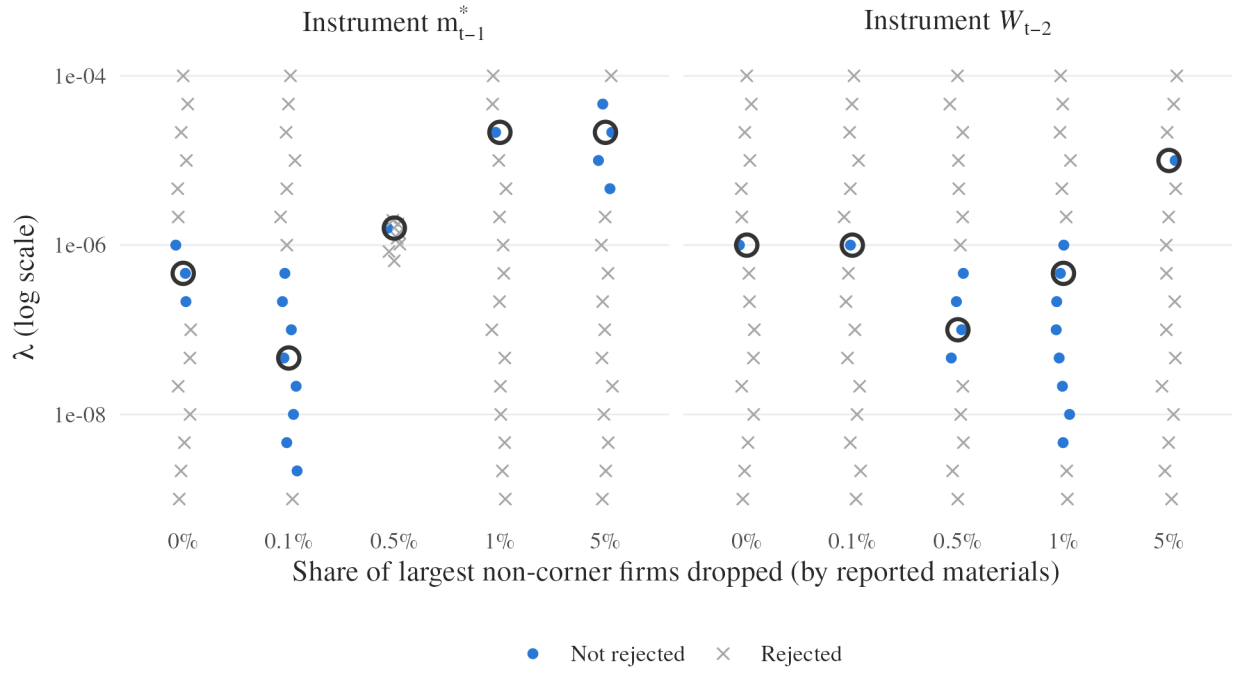


Figure 7: Trim-cutoff test inversion, instruments m_{t-1}^* and $\mathcal{W}_{t-2}$. At each trim level (share of the largest non-corner firms dropped, by reported materials), every tested λ is marked as not rejected (dot) or rejected (cross); the ring marks the grid minimum within that trim level. Pass/fail uses $2n(\hat{L}_n - \min \hat{L}_n)$ against $\chi_{8,.95}^2$, from the earlier eight-moment system.

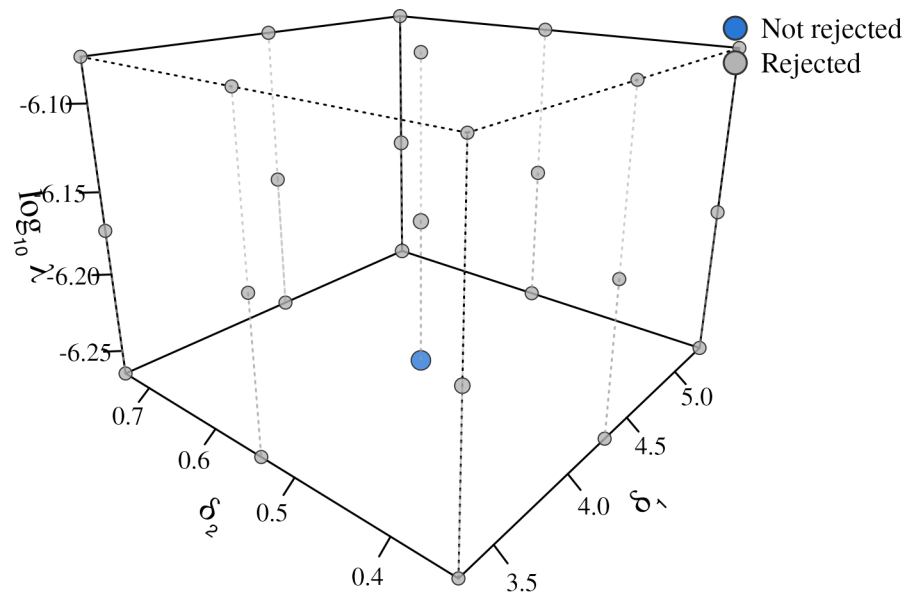


Figure 8: Joint $(\lambda, \delta_1, \delta_2)$ cube, instrument m_{t-1}^* : 27 tested cells, with δ_0 and γ profiled out at each. Blue: not rejected by the conservative test ($\chi^2_{9,.95}$); grey: rejected. Ball size grows with the test's p-value.

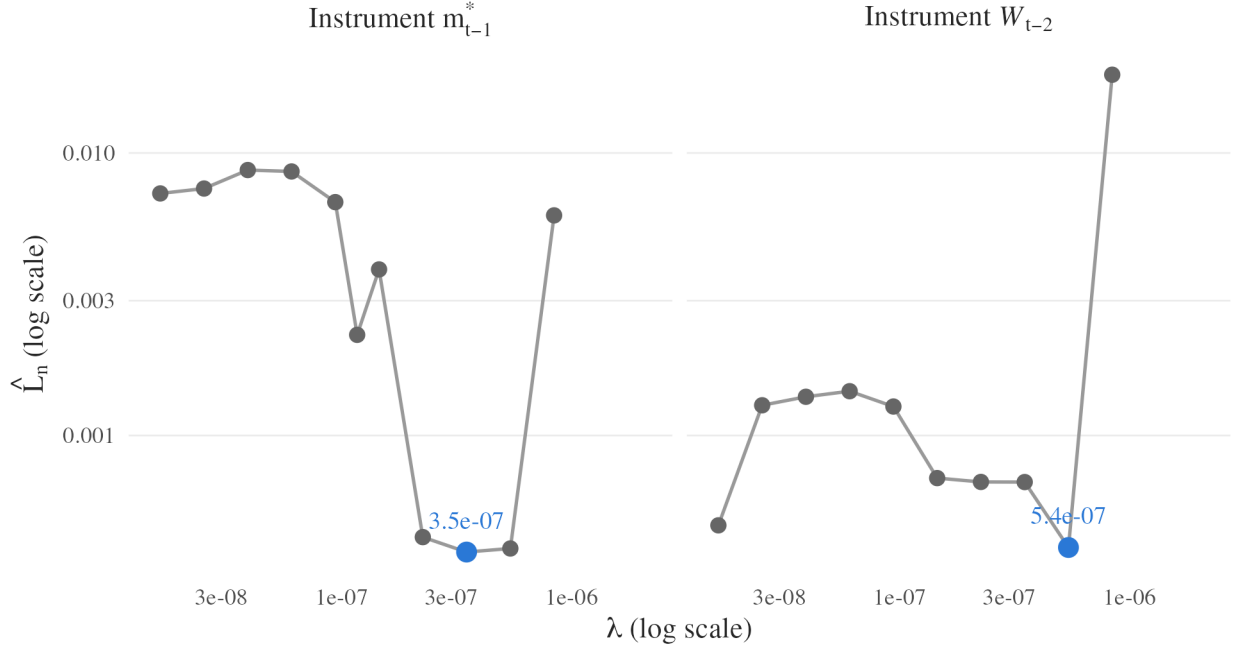


Figure 9: λ -grid: the CUE objective $\hat{L}_n$ at each tested λ , with $\delta_0, \delta_1, \delta_2$ and γ profiled out, for instruments m_{t-1}^* and W_{t-2} . The labelled point is each grid's minimum.

A.4 Test inversion: the conservative test, and why

- **Source definition (Schennach 2014, Definition F.1 and Theorem F.1):** with $\hat{g}(\theta, \gamma) = \frac{1}{n} \sum_{i=1}^n \tilde{g}(Z_i, \theta, \gamma)$ (the sample average of the tilted moment from Equation 32) and $\hat{V}(\theta, \gamma) = \frac{1}{n} \sum_{i=1}^n \tilde{g}(Z_i, \theta, \gamma) \tilde{g}(Z_i, \theta, \gamma)'$, her continuous updating objective is $\hat{L}_n^S(\theta) = \sup_{\gamma \in \mathbb{R}^{d_g}} -\frac{1}{2} \hat{g}(\theta, \gamma)' \hat{V}(\theta, \gamma)^- \hat{g}(\theta, \gamma)$, with $\hat{V}^-$ a generalized inverse. Theorem F.1: at any θ satisfying the moment condition, $-2n\hat{L}_n^S(\theta)$ is asymptotically bounded above by $\chi_{d_g}^2$. Maximizing $\hat{L}_n^S$ over γ is the same as minimizing the nonnegative quadratic form $\frac{1}{2} \hat{g}' \hat{V}^- \hat{g}$; we report this positive quantity as $\hat{L}_n(\theta)$, so $TS(\theta) = 2n\hat{L}_n(\theta) = -2n\hat{L}_n^S(\theta)$ — same statistic, sign flipped purely so the optimizer minimizes a nonnegative objective instead of maximizing a nonpositive one. $\hat{V}^-$ in our implementation discards eigen-directions with eigenvalues below 10^{-8} times the largest, for numerical stability.
- **Conservative test** (Theorem F.1, absolute): $TS_{\text{cons}} = 2n\hat{L}_n$ against $\chi_{d_g, .95}^2$ directly — no credit for profiling out nuisance parameters or γ .
- **Why the conservative test:** the counterfactual's purpose is to support a policy claim, so the confidence sets use the test whose size is guaranteed not to exceed its nominal level.
- Full derivations: Schennach (2014), Theorem F.1 and its proof; .

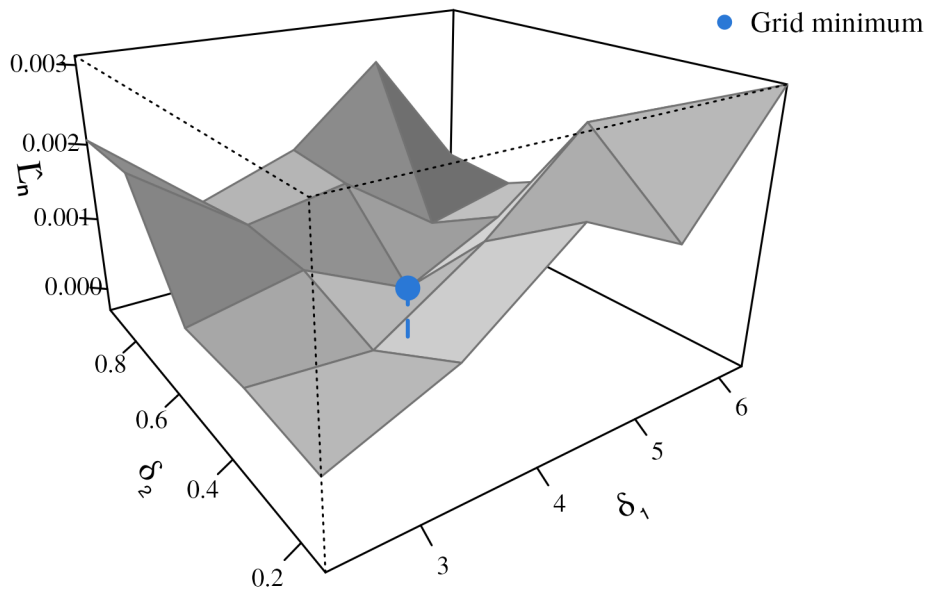


Figure 10: (δ_1, δ_2) -grid: $\hat{L}_n$ at each tested pair, with λ , δ_0 and γ profiled out, instrument m_{t-1}^* . The marked point is the grid minimum, $(\delta_1, \delta_2) = (4.3, 0.54)$.

A.5 Back-of-envelope: revenue loss and its tax-rate elasticity

Revenue lost to undetected evasion:

- Mean $\text{Loss}_i = \tau_{P,i}(1 - q(e_i))e_i$ (the extra credit paid out that perfect detection would have caught) across all firms: **\$53.32 real COP per firm-period; \$1,718,598 total** across the sample.
- **Reported as a share of *potential* revenue, the standard VAT-gap convention** – not of actual collections. Potential = actual + loss = 1702.24 + 53.32 = \$1755.56; gap = 1 – actual/potential = **3.04%**.
- **A narrow-channel number:** one mechanism (materials-credit overreporting), one country, one structural sample – smaller than the official comparators’ *non-compliance-only* gap shares (14.1 percent Colombia, 30.7 percent Chile, 54.4 percent Mexico, all CIAT, as % of potential), which capture every form of non-compliance, not just this channel.

Revenue elasticity of the tax rate:

- Closed form from the same $\tilde{e}(\Delta)$ used throughout this chapter: $\frac{\partial R_i}{\partial x}\bigg|_{x=1} = -\tau_{P,i} \left[M_i - e_i + \lambda e_i^2 + \frac{1}{2\lambda} \right]$, $x \equiv 1 + \Delta$.
- **Two one-sided elasticities, not one** – the same kink at $\Delta = 0$ documented in Section 6.4 makes the two-sided derivative ill-defined: $\varepsilon_{\text{right}} = -20.30$, $\varepsilon_{\text{left}} = -9.68$ (both aggregate, ratio of means, $x = 1$). Both large; the asymmetry is the same mechanism as the Results section’s own asymmetric-response finding, not a new one.
- **Elasticity “with respect to the tax rate” is the same number, not a third one, and this is exact, not approximate.** The model only ever moves τ_P through the proportional shifter x ($\tilde{\tau}_{P,i} = x\tau_{P,i}$, $\tau_{P,i}$ fixed), so $d \ln \tilde{\tau}_{P,i} = d \ln x$ identically at every firm, every x – not just at $\Delta = 0$.
- Median interior-firm elasticity, as a distinct “typical firm” measure: -195.99 – far larger in magnitude because the median evading firm’s own R_i is much smaller than the sample mean.

B The Single-Tax Model and the Second-Order Condition

This appendix presents the single-tax model that Section 1 nests, and derives the second-order condition (SOC) for the firm’s problem, which translates into the curvature restriction on the detection function $q(e)$ used throughout the paper.

B.1 The single-tax model

The single-tax model keeps every assumption of Section 1 except one: a single rate τ applies to both sales and purchases. The firm solves

$$\begin{aligned} \max_{K,L,M,e} \mathbb{E}[\pi_{it}] = & (1 - \tau)(P_t K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it}) - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \\ & + [1 - q(e_{it})]\tau \rho_t e_{it} - q(e_{it})\phi \tau \rho_t e_{it} - \kappa_{it}(e_{it}, \omega_{it}), \end{aligned} \quad (34)$$

which is Equation 2 with $\tau_S = \tau_P = \tau$. The tax rate drops out of the materials first-order condition,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta-1} \exp(\omega_{it}) = \rho_t / P_t, \quad (35)$$

so the gross log share, $\ln(\rho_t M_{it} / P_t Y_{it}) = \ln \beta - \varepsilon_{it}$, takes the place of the net share in Equation 4. The evasion first-order condition is Equation 5 with τ in place of τ_P ,

$$\tau \rho_t [1 - (1 + \phi)(q(e_{it}) + q'(e_{it})e_{it})] = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}}, \quad (36)$$

and firms for which $\tau \rho_t \leq \partial \kappa_{it}(0, \omega_{it}) / \partial e_{it}$ do not overreport.

B.2 The second-order condition

B.2.1 Setup

Recall the firm's problem in the two-tax model, Equation 2, and its two first-order conditions, Equation 3 and Equation 5. For a candidate solution $(K_{it}, L_{it}, M_{it}, e_{it})$ to be a maximum, the Hessian of expected profit must be negative semi-definite at that point.

B.2.2 The Hessian is block-diagonal

Two features of the model make the Hessian easy to sign.

First, $M_{it} \perp e_{it}$: neither the materials FOC (Equation 3) nor the evasion FOC (Equation 5) contains the other's choice variable, so every cross-partial between e_{it} and (K_{it}, L_{it}, M_{it}) is zero. Second, the production function $Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it})$ is Cobb-Douglas, hence strictly concave in

(K_{it}, L_{it}, M_{it}) jointly. The Hessian is therefore block-diagonal,

$$\nabla^2 \mathbb{E}[\pi_{it}] = \begin{bmatrix} \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial e_{it}^2} & 0 & 0 & 0 \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial M_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial M_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial M_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial L_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial L_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial L_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial K_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial K_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial K_{it}^2} \end{bmatrix} \quad (37)$$

with a single nonzero entry in the first row and column. Negative definiteness requires the leading principal minors to alternate in sign (odd-order minors negative, even-order positive). Since the lower 3×3 block is already negative definite (strict concavity of the Cobb-Douglas production function in K_{it}, L_{it}, M_{it}), the *only* condition the evasion margin adds to the firm's problem is that the first leading principal minor — the top-left scalar entry — is itself negative:

$$\frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial e_{it}^2} < 0.$$

The capital, labour, and materials choices place no restriction on the evasion technology, and vice versa; the SOC for e_{it} can be analyzed entirely on its own.

B.2.3 The scalar condition

Differentiating the evasion FOC (Equation 5) once more with respect to e_{it} ,

$$\frac{\partial^2 \mathbb{E}[\pi_{it}]}{\partial e_{it}^2} = -\tau_P \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) - \frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} < 0. \quad (38)$$

Rearranging, the SOC holds whenever

$$\tau_P \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) + \frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} > 0,$$

a sum of two terms. It is therefore *sufficient* — not necessary — that either term alone be strictly positive:

$$\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} \geq 0 \quad \text{or} \quad \tau_P \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) \geq 0. \quad (39)$$

Under this paper's headline cost function, $\kappa_{it}(e_{it}, \omega_{it}) = e_{it} \exp\{\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it}\}$

— linear in e_{it} — the first term is identically zero, $\partial^2 \kappa_{it} / \partial e_{it}^2 = 0$. The cost function therefore contributes nothing toward satisfying the SOC on its own; the entire burden falls on the second condition, the curvature of the detection function.

B.2.4 The curvature restriction

Since $\tau_P \rho_t (1 + \phi) > 0$, the second condition in Equation 39 reduces to $2q'(e_{it}) + e_{it}q''(e_{it}) \geq 0$. Economic theory imposes $q'(e_{it}) > 0$ (detection is increasing in overreporting) and $e_{it} > 0$ for an interior evader, so dividing through by $q'(e_{it})e_{it} > 0$ gives

$$-\frac{q''(e_{it})}{q'(e_{it})} e_{it} < 2. \quad (40)$$

This is a restriction on the **curvature of q alone** — a firm's evasion cost never enters, given the linear-in- e specification above. It says q' may decline as e_{it} rises (a concave detection function), but not too fast relative to its own level: the semi-elasticity of the marginal detection probability with respect to e_{it} must exceed -2 .

B.2.5 Application to the three candidate detection functions

Table 20 applies Equation 40 to each of the three functional forms considered in Section 1.

Table 20: Second-order condition by functional form of $q(\cdot)$

	$q(e)$	$q''(e)$	$-q''e/q'$	SOC holds when
Linear	$\min\{\lambda e, 1\}$	0	0	always
Concave	$1 - e^{-\lambda e}$	$-\lambda^2 e^{-\lambda e}$	λe	$\lambda e < 2$
Convex	$\min\{(\lambda e)^k, 1\},$ $k > 1$	$k(k-1)\lambda^2(\lambda e)^{k-2}$	$1 - k$	always ($k > 1 \Rightarrow$ $1 - k < 0 < 2$)

Linear detection, $q(e) = \min\{\lambda e, 1\}$ (headline specification): $q''(e) = 0$ on the interior, so $-q''e/q' = 0 < 2$ unconditionally — the SOC holds at every interior point, with no restriction on λe beyond the FOC's own ceiling ($\lambda e < 1/2$, from Section 1).

Concave detection, $q(e) = 1 - e^{-\lambda e}$: $q'(e) = \lambda e^{-\lambda e}$ and $q''(e) = -\lambda^2 e^{-\lambda e}$, so $-q''e/q' = \lambda e$. The SOC requires $\lambda e < 2$ — a genuine restriction, distinct from (and, for this functional form, looser than) the FOC's own ceiling $\lambda e < 1$ derived in Section 1. Since $\lambda e < 1 \Rightarrow \lambda e < 2$, the FOC ceiling is the binding constraint in practice; the SOC is automatically satisfied wherever the FOC has an interior solution at all.

Convex detection, $q(e) = \min\{(\lambda e)^k, 1\}$, $k > 1$: $q'(e) = k\lambda(\lambda e)^{k-1}$ and $q''(e) = k(k-1)\lambda^2(\lambda e)^{k-2}$, so $-q''e/q' = 1 - k$. Since $k > 1$, this is always negative, hence always less than 2 — convex detection satisfies the SOC unconditionally, for any $k > 1$.

Summary. All three candidate detection functions satisfy the second-order condition on the relevant part of their domain — linear and convex unconditionally, concave provided $\lambda e < 2$, which is already implied by that function's own FOC-ceiling. The choice among them is therefore not disciplined by the SOC; it is a modelling choice about the detection technology, addressed on economic grounds in Section 6.

The derivation carries over to the single-tax model (Section B.1) with τ in place of τ_P : the rate scales the curvature term but does not change its sign, so Equation 40 is the same condition in both models.

C Production-Function Estimates for All Industries

The stage-2 estimation in Section 6 uses every industry, not only the seven reported in Section 4. Table 21 lists the production-function estimates it uses, with instrument $\tilde{\mathcal{W}}_{it-2}$.

D Robustness of the Test for Overreporting

Table 5 reports the sharp version of the test, which takes $\ln \hat{\beta}$ from corporations as the true value. This appendix reports two alternatives. Table 22 gives the conservative test, which gives no credit for the corporations' moment. Table 23 gives a bootstrap test that re-estimates $\ln \hat{\beta}$ in every replication, so it carries the sampling error of the first stage.

E The Tax Wedge in the Reform Estimates

A tax change can move reported materials shares mechanically, without any change in overreporting: prices and the tax paid on sales and purchases adjust to the new rates. The two-tax model isolates the part of this adjustment that runs through the sales-tax rates. With different rates on sales, τ_S , and on purchases, τ_P , the gross materials share carries the wedge $\ln((1 - \tau_S)/(1 - \tau_P))$, which depends on the rates themselves. The 1983 reform changed those rates in the liable industries, so a rise in the gross share after the reform could reflect either more overreporting or this wedge. The event study in Section 5 uses the share net of sales taxes, which removes the wedge.

Table 24 estimates the wedge's contribution directly: the same event study, with the gross minus the net share as the outcome. Its coefficients equal the gap between the event study on the gross share and on the net share. Before the reform the wedge does not move. After it, the wedge rises by

Table 21: Production-function estimates by industry, instrument $\tilde{\mathcal{W}}_{it-2}$, with 95 percent test-inversion regions (sharp test).

Industry	$\hat{\beta}$	β , sharp	$\hat{\alpha}_K$	α_K , sharp	$\hat{\alpha}_L$	α_L , sharp	n
311 Food products	0.592	[0.562, 0.623]	0.229	[0.185, 0.270]	0.202	[0.155, 0.250]	5,825
312 Other food products	0.574	[0.495, 0.667]	0.202	[0.135, 0.280]	0.283	[0.185, 0.375]	1,168
313 Beverages	0.287	[0.269, 0.306]	0.406	[0.295, 0.575]	0.307	($\cdot$, 0.520]	969
314 Tobacco	0.265	[0.201, 0.350]	0.588	($\cdot$, $\cdot$)	0.132	($\cdot$, 0.600]	127
321 Textiles	0.385	[0.357, 0.416]	0.300	[0.210, 0.375]	0.349	[0.230, 0.485]	2,693
322 Wearing apparel	0.335	[0.288, 0.391]	0.182	[0.085, 0.265]	0.392	[0.320, 0.470]	4,247
323 Leather products	0.553	[0.506, 0.606]	0.236	[0.140, 0.345]	0.321	[0.185, 0.440]	559
324 Footwear	0.412	[0.365, 0.465]	0.119	($\cdot$, 0.270]	0.427	[0.190, 0.570]	1,063
331 Wood products	0.324	[0.263, 0.398]	0.076	($\cdot$, 0.200]	0.554	[0.400, 0.700]	951
332 Furniture	0.348	[0.267, 0.454]	0.176	[0.060, 0.270]	0.547	[0.410, 0.725]	985
341 Paper products	0.501	[0.453, 0.554]	0.095	($\cdot$, 0.340]	0.344	[0.045, 0.620]	1,062
342 Printing and publishing	0.333	[0.294, 0.377]	0.292	[0.215, 0.360]	0.408	[0.305, 0.530]	1,967
351 Industrial chemicals	0.340	[0.292, 0.396]	0.308	($\cdot$, 0.495]	0.362	[0.005, 0.665]	759
352 Other chemicals	0.407	[0.379, 0.437]	0.325	[0.240, 0.420]	0.273	[0.165, 0.400]	2,205
354 Petroleum and coal [§]	0.603	[0.598, 0.609]	0.172	($\cdot$, 0.525]	0.160	($\cdot$, 0.755]	152
355 Rubber products	0.415	[0.383, 0.450]	0.244	($\cdot$, 0.505]	0.330	[0.140, 0.560]	522
356 Plastic products	0.488	[0.455, 0.523]	0.286	[0.210, 0.350]	0.213	[0.130, 0.305]	1,625
361 Pottery and china	0.239	[0.214, 0.267]	0.030	($\cdot$, 0.885]	0.675	($\cdot$, $\cdot$)	211
362 Glass products	0.271	[0.223, 0.329]	0.338	[0.185, 0.510]	0.352	[0.030, 0.655]	380
369 Non-metallic minerals	0.244	[0.207, 0.288]	0.333	[0.210, 0.485]	0.549	[0.210, 0.740]	1,480
371 Iron and steel	0.381	[0.297, 0.490]	0.116	($\cdot$, 0.355]	0.503	[0.075, 0.945]	326
372 Non-ferrous metals	0.498	[0.457, 0.542]	0.222	($\cdot$, 0.715]	0.258	($\cdot$, 0.510]	215
381 Metal products	0.400	[0.369, 0.434]	0.234	[0.160, 0.310]	0.391	[0.295, 0.495]	3,646
382 Non-electrical machinery	0.374	[0.340, 0.412]	0.225	[0.100, 0.350]	0.438	[0.320, 0.575]	2,139
383 Electrical machinery	0.395	[0.352, 0.443]	0.226	[0.055, 0.900]	0.388	($\cdot$, 0.630]	1,381
384 Transport equipment	0.431	[0.375, 0.496]	0.220	[0.145, 0.310]	0.479	[0.345, 0.590]	1,427
385 Professional equipment	0.334	[0.293, 0.381]	0.183	($\cdot$, 0.365]	0.551	[0.295, 0.715]	416
390 Other manufacturing	0.333	[0.271, 0.409]	0.209	[0.115, 0.335]	0.501	[0.325, 0.690]	979

Output elasticities of materials ($\hat{\beta}$, from the first stage on corporations, log materials share net of sales taxes), capital ($\hat{\alpha}_K$) and labour ($\hat{\alpha}_L$) for every industry used in the stage-2 estimation of the detection and evasion-cost parameters. Instrument $\tilde{\mathcal{W}}_{it-2}$. Point estimates: the minimum of the test-inversion statistic, with the covariance of the moments re-estimated at each candidate (α_K , α_L); β , sharp: 95% test-inversion region for β from the corporations' share moment ($\chi^2_{1,0.95}$; the production-function moments are exactly identified given β and profiled). α , sharp: projections of the 95% test-inversion region ($\chi^2_{2,0.95}$, credit for profiling γ_0 , γ_1) on a 0.005 grid over $[0, 1]^2$. ($\cdot$ or $\cdot$): the bound of the parameter space (0 or 1) is not rejected. n : observations in the first stage. Juridical organization codes 6–9 excluded; industry 353 (petroleum refineries) has three observations left and is not estimated. [§] Two corporate plants: the plant-clustered variance behind the region for β is unreliable.

Table 22: Test for overreporting of materials by industry, conservative test. Mean of $\mathcal{V}_{it}$ among unincorporated firms and 95 percent test-inversion region for its expectation, with $\ln \beta$ estimated from corporations.

Industry	$\hat{\mu}$	95% region, conservative
311 Food products	0.024	($\cdot$, 0.054]
312 Other food products	-0.018	($\cdot$, 0.100]
313 Beverages	0.390	[0.287, 0.492]
321 Textiles	0.151	[0.105, 0.197]
322 Wearing apparel	0.224	[0.187, 0.261]
323 Leather products	-0.192	†
324 Footwear	0.095	[0.050, 0.140]
331 Wood products	0.250	[0.156, 0.344]
332 Furniture	-0.007	($\cdot$, 0.075]
341 Paper products	-0.066	($\cdot$, 0.022]
342 Printing and publishing	0.074	[0.026, 0.122]
351 Industrial chemicals	0.165	($\cdot$, 0.375]
352 Other chemicals	0.057	($\cdot$, 0.125]
356 Plastic products	-0.107	†
369 Non-metallic minerals	0.193	[0.087, 0.298]
381 Metal products	-0.005	($\cdot$, 0.033]
382 Non-electrical machinery	-0.014	($\cdot$, 0.036]
383 Electrical machinery	0.034	($\cdot$, 0.103]
384 Transport equipment	-0.129	†
390 Other manufacturing	0.057	($\cdot$, 0.130]

$\hat{\mu}$: mean of $\mathcal{V}_{it} = \ln(\rho_t M_{it}^* / P_t Y_{it}) - \ln \hat{D}$ among unincorporated firms, with the log materials share net of sales taxes and $\ln \hat{D}$ the mean among corporations. 95% test-inversion region for $\mu = E[\mathcal{V}_{it}]$ over the grid $\mu \in [0, 1]$ (under the model $\mu \geq 0$), taking $\ln \hat{D}$ as the true value; plant-clustered, efficiently weighted moments; conservative, $\chi_{2,0.95}^2 = 5.99$ critical value (no credit for the corporations' moment). Where the region excludes 0, the test rejects no overreporting. ($\cdot$, b): $\mu = 0$ is not rejected. † Empty region: every $\mu \in [0, 1]$ is rejected.

Table 23: Bootstrap test for overreporting of materials by industry. Mean of $\mathcal{V}_{it}$ among unincorporated firms, with $\ln \beta$ re-estimated from corporations in every replication.

Industry	$\hat{\mu}$	95% CI
311 Food products	0.024	[−0.035, 0.081]
312 Other food products	−0.018	[−0.215, 0.148]
313 Beverages	0.390***	[0.288, 0.485]
321 Textiles	0.151***	[0.063, 0.236]
322 Wearing apparel	0.224***	[0.056, 0.374]
323 Leather products	−0.192	[−0.305, −0.087]
324 Footwear	0.095*	[−0.021, 0.232]
331 Wood products	0.250**	[0.014, 0.461]
332 Furniture	−0.007	[−0.277, 0.261]
341 Paper products	−0.066	[−0.190, 0.051]
342 Printing and publishing	0.074	[−0.061, 0.203]
351 Industrial chemicals	0.165*	[−0.050, 0.385]
352 Other chemicals	0.057	[−0.029, 0.150]
356 Plastic products	−0.107	[−0.195, −0.025]
369 Non-metallic minerals	0.193**	[0.005, 0.376]
381 Metal products	−0.005	[−0.095, 0.079]
382 Non-electrical machinery	−0.014	[−0.125, 0.089]
383 Electrical machinery	0.034	[−0.089, 0.161]
384 Transport equipment	−0.129	[−0.279, 0.021]
390 Other manufacturing	0.057	[−0.148, 0.282]

$\hat{\mu}$: mean of $\mathcal{V}_{it}$ among unincorporated firms, $\ln \hat{D}$ the mean among corporations. Bootstrap with 2,000 replications: plants resampled separately within corporations and within unincorporated firms, with $\ln \hat{D}$ re-estimated in every replication, so the test carries the first-stage sampling error. Basic (pivotal) 95% confidence intervals. *, **, *** denote significance at the 10%, 5%, and 1% levels (one-sided). The intervals are two-sided, so a one-sided rejection at 5% can coexist with an interval that includes zero.

Table 24: The tax wedge in the event study of the 1983 reform: gross minus net-of-sales-tax log materials share, difference relative to 1983.

Year	Unincorporated firms			
	Liable	Exempt	LLCs, liable	Proprietorships, liable
1981	0.000 (0.001)	-0.003*** (0.001)	0.000 (0.001)	0.001 (0.003)
1982	0.001 (0.001)	0.002*** (0.001)	0.001 (0.001)	0.002 (0.003)
1983	—	—	—	—
1984	0.007*** (0.001)	0.002*** (0.001)	0.006*** (0.001)	0.013*** (0.003)
1985	0.008*** (0.001)	0.003*** (0.001)	0.007*** (0.001)	0.013*** (0.004)
1986	0.007*** (0.001)	0.003*** (0.001)	0.007*** (0.002)	0.011*** (0.004)
1987	0.006*** (0.001)	0.005*** (0.001)	0.005*** (0.001)	0.008** (0.003)
1988	0.004*** (0.001)	0.004*** (0.001)	0.003* (0.001)	0.009*** (0.003)
1989	0.006*** (0.001)	0.004*** (0.001)	0.005*** (0.001)	0.008** (0.004)
1990	0.003** (0.001)	0.004*** (0.001)	0.002 (0.001)	0.008** (0.004)
1991	0.010*** (0.001)	0.002*** (0.001)	0.010*** (0.002)	0.012*** (0.004)

Outcome: the gross minus the net-of-sales-tax log materials share, $\ln(\rho_t M_{it}^* / P_t Y_{it}) - \ln((1 - \tau_{P,it}) \rho_t M_{it}^* / (P_t Y_{it} - \tau_{S,it} \text{sales}_{it}))$, the tax wedge that the gross share carries. Same specification as the differences in the event study of the 1983 reform: difference relative to 1983 (reference, —), industry fixed effects, unincorporated firms (or LLCs, proprietorships) relative to corporations, in sales-tax-labile and exempt industries. The coefficients equal the gap between the event study on the gross share and on the net share. Standard errors in parentheses, clustered by plant. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Observations: 39,495; juridical organization codes 6–9 excluded.

about 0.006 to 0.008 for unincorporated firms in liable industries, roughly a tenth of their measured response, and by 0.011 to 0.013 for proprietorships in 1984–1986, most of their response in those years. An event study on the gross share would attribute this mechanical component to overreporting.

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